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MILITARY STANDARD

TYPES OF ENGINEERING DRAWINGS

DEFINITIONS AND REQUIREMENTS

FSC - DRPR

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DEFENSE SUPPLY AGENCY
Cameron Station
Alexandria, Va.

Types of Engineering Drawings Definitions and Requirements

MIL-STD-7A

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, effective immediately.

2. Recommended corrections, additions, or deletions should be addressed to Standardization Division, Directorate of Logistics Services, Defense Supply Agency, Cameron Station, Alexandria, Va.

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MILITARY STANDARD

TYPES OF ENGINEERING DRAWINGS DEFINITIONS AND REQUIREMENTS

1. SCOPE

1.1 This standard describes the basic types of engineering drawings prepared by or for the Departments of the Army, the Navy and the Air Force.

1.2 General requirements for engineering drawings are specified in other documents. Basic characteristics applicable to individual types of engineering drawings are set forth herein. The figures herein are for illustration of text only and contain no mandatory requirements.

2. REFERENCED DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this standard to the extent specified herein:

MILITARY STANDARDS

MIL-STD-34 -- Preparation of Drawings for Optical Elements and Optical Systems; General Requirements for
MIL-STD-275 - Printed Wiring for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS ASSOCIATION

Y14.15 -- Electrical Diagrams

(Application for copies should be addressed to the American Standards Association Incorporated, 10 East 40th Street, New York 16, N. Y.)

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3. Definitions and Requirements.

3.1 General definition.

3.1.1 Engineering drawing. An engineering drawing is a document that discloses by means of pictorial or textual presentations, or combinations thereof, the physical or functional engineering requirements of an item.

3.2 Arrangement drawing.

3.2.1 Definition. An arrangement drawing is an outline drawing that depicts in any projection or perspective, with or without controlling dimensions, the relationship of major units of the item depicted (see figure 1).

3.2.2 Requirements. An arrangement drawing shall show sufficient views of the item so that a general understanding is conveyed of the configuration and location of major units. Overall, locating, and other general dimensions necessary to define the configuration may be shown. Major units shall be identified.

3.3 Assembly drawing.

3.3.1 Definition. An assembly drawing depicts the assembled relationship of (a) two or more parts, or (b) a combination of parts and subordinate assemblies, or (c) a group of assemblies required to form an assembly of higher order (see figures 2 and 3).

3.3.2 Requirements. An assembly drawing shall contain sufficient views to show the relationship between each subordinate assembly and part comprising the assembly depicted. Subordinate assemblies and parts shall be called out in the field of the drawing, either by their identifying (part) numbers or by find (item) numbers cross-referenced to the identifying numbers in a table, parts list, or list of material. When information regarding the assembled relationship and identification of parts is shown on assembly drawings of subordinate assemblies, it should not be repeated on the assembly drawing of next higher order, only the identifying number of each such subordinate assembly, its configuration and location being shown. Assembly drawings shall contain references to pertinent installation drawings, wiring and schematic diagrams, etc., as applicable. The division of an item into subordinate assemblies should be in accordance with practical assembly and disassembly procedures.

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3.3.2.1 Electrical items. Electrical items shall be shown and identified on assembly drawings when they are mechanically mounted; however, small electrical items which are mounted by means of wire connectors affixed thereto may be shown and identified either on the assembly drawing or on the pertinent wiring diagram.

3.3.2.2 Attaching parts. Attaching parts (bolts, nuts, washers, etc.) required to mount and retain assemblies on foundations or on assemblies of higher order shall be called out on the drawing showing the item on which the attachment takes place.

3.3.3 Wiring harness drawing.

3.3.3.1 Definition. A wiring harness drawing shows the path of a group of wires laced together in a specified configuration, so formed to simplify installation (see figure 4).

3.3.3.2 Requirements. The drawing shall show all dimensions necessary to define the harness form and termination points; or a grid system, or vertical and horizontal graphic scales may be provided, in lieu of dimensions, for use in ascertaining the required dimensional data. The drawing shall also include a wire data tabulation of wire numbers, circuit reference designations, color codes, lengths, material specifications, and other data, as necessary. Included in note form should be instructions, or references thereto, for the preparation and installation of the harness, associated schematic diagram, and the wiring diagram.

3.4 Book form drawing.

3.4.1 Definition. A book form drawing is an assemblage of drawings and related data disclosing by means of pictorial delineations or technical tabulations, or combinations thereof, the engineering requirements of an item, a family of items, or a system. A book form drawing is used for special purpose applications in which it is expeditious to provide a document consisting of numerous small sheets, suitable for binding into book form (see figures 5(a) through 5(e)).

3.4.2 Requirements. A book form drawing should be prepared preferably on A-size sheets. Other standard size sheets may be used provided the final original document sheet sizes are reduced to 11 inch height and can be folded to 8.5 inch width, and legibility when so reduced is maintained. Book form drawings shall not be prepared to circumvent the requirements for furnishing the individual types of drawings required for the delineation of an item.

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3.4.3 Running list

3.4.3.1 Definition. A running list is a book form drawing consisting of tabular data and instructions required to establish wiring connections between units of an equipment, or between equipments, sets or assemblies of a system. A running list is a form of interconnection diagram (see paragraph 3.11.1.1(d) and figure 5).

3.4.3.2 Requirements. Normally, the principal sections of a running list shall be as follows:

(a) Title sheet
(b) Revision status of sheets tabulation
(c) Table of contents
(d) Referenced documents
(e) Illustrations. The figures necessary to supplement the tabulations shall be provided.
(f) List of units. Units shall be listed by unit designation and nomenclature.
(g) Summaries of cabling information by units
(h) Cable running sheets. For each cable, an individual sheet shall show the terminals to which each wire in the cable is attached, the color code of each wire, and the Government procuring activity cable number (if assigned).

3.5 Construction drawing.

3.5.1 Definition. A construction drawing delineates the design of buildings, structures, or related construction, ashore or afloat, individually or in groups. Construction drawings in general, present design information by pictorial plans, elevations, sections and details. The plan drawing covered in paragraph 3.23 is one type of construction drawing.

3.5.2 Requirements. A construction drawing, or set of construction drawings, shall establish all the interrelated elements of the design, including pertinent services, equipment, utilities and other engineering details. The requirements for a plan drawing in paragraph 3.23.2 also apply to a set of construction drawings.

3.5.3 Exceptions in regard to construction drawings. Maps (except those accompanying or used in conjunction with construction drawings), sketches, presentation drawings, perspectives and renderings are not construed to be construction drawings within the meaning of this standard.

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3.6 Control drawing.

3.6.1 Definition. A control drawing is a procurement document for a commercial or vendor item specifying therefor engineering requirements to the extent necessary to ensure identification and reprocurement of the item(s) delineated, normally without disclosing the details of design. Control drawings are designated as either (a) Source control drawings, or (b) Specification control drawings.

3.6.2 Requirements. A control drawing shall disclose, as applicable, configuration, dimensions of envelope (expressed as limits or as maxima), mounting and mating dimensional limits, and all other interface characteristics that affect the physical interchangeability of the item. Additionally, performance, reliability, maintainability, environmental and other requirements, as necessary, to meet design requirements, and test requirements to the extent necessary to ensure that such design requirements have been met, shall also be specified, either on the drawing or in a document invoked thereon. If the item(s) delineated contains an electrical or electronic (or other engineering) circuit, a schematic, connection, or other appropriate diagram, and any other information necessary to establish external connections, shall be included on the drawing, or in a referenced document. Also, if the item delineated is a repairable assembly for which replacement parts are provisioned, a listing of such parts, for each vendor named, shall be listed (part identification, nomenclature, and, as applicable, documentation number) on the drawing or in a referenced list.

3.6.2.1 Source control drawing. A source control drawing restricts procurement (1) to the specified item(s) described on the drawing and (2) to the stated source(s) of supply designated thereon. These restrictions are imposed by the cognizant design activity to ensure procurement of the only item(s) known, as a result of test or evaluation, to be satisfactory for the stated critical application. The drawing shall include the following note: "ONLY THE ITEM DESCRIBED ON THIS DRAWING WHEN PROCURED FROM THE VENDOR(S) LISTED HEREON IS APPROVED BY (name, and address of cognizant design activity) FOR USE IN THE APPLICATION(S) SPECIFIED HEREON. A SUBSTITUTE ITEM SHALL NOT BE USED WITHOUT PRIOR TESTING AND APPROVAL BY (name of cognizant design activity) OR BY (name of Government procuring activity)". The drawing shall also include, under the heading "APPROVED SOURCE(S) OF SUPPLY", a list showing each approved vendor (name, address, and vendor's item identification) and the specific application involved. In addition, the notation, "SOURCE CONTROL DRAWING" shall be shown, either adjacent to or as a subtitle in the title block. The item(s) thus disclosed shall be called out on the assembly drawing(s) and the associated list(s) by the source control drawing number (and dash number, if any). When essential to ensure item identification, the source control drawing number shall be marked on the item. The vendor's item identification may also be marked on the item in conjunction with the source control drawing number. Subsequent to drawing release, whenever another vendor's item is qualified for the specific application, or when a new application is found for the item delineated, the drawing may be revised, rather than a new drawing issued, to show the new vendor or application (see figure 6).

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3.6.2.2 Specification control drawing. A specification control drawing delineates a commercial or vendor item for which all of the engineering and test requirements specified can be met in commercial channels without imposing restrictions essential in source control procurement. The drawing, under the heading "SUGGESTED SOURCES OF SUPPLY", shall list each vendor (name, address, and the vendor's item identification) whose item(s) has been accepted as meeting all drawing requirements. Additionally, the notation "SPECIFICATION CONTROL DRAWING" shall appear, either adjacent to or as a subtitle in the title block. The item(s) thus disclosed shall be called out on the assembly drawing(s) and the associated list(s) by the specification control drawing number (and dash number, if any). When practicable, the vendor's item identification shall be marked on the item. A specification control drawing shall not be prepared for a commercial or vendor item which has already been satisfactorily described and identified in an Industry Association or Government specification or standard (see figure 7).

3.7 Coordination drawing.

3.7.1 Definition. A coordination drawing discloses the physical and functional interface engineering requirements of a subsystem which affect the design or operation of cofunctioning subsystems. Coordination drawings are used as authoritative design control documents, delineating subsystem interface engineering data which have been coordinated for the purpose of (a) establishing and maintaining compatibility between cofunctioning subsystems; (b) controlling interface designs thereby preventing inadvertent changes to subsystems requirements which would affect compatibility with cofunctioning subsystems; (c) communicating design decisions and changes to participating activities (see figure 8).

3.7.2 Requirements. A coordination drawing shall delineate, as necessary, (a) configuration and all interface dimensional data applicable to the envelope, mounting and mating of the subsystem; (b) complete interface engineering requirements, such as mechanical, electrical, electronic, hydraulic, pneumatic, optical, etc., which affect the physical or functional characteristics of cofunctioning subsystems; and (c) any other characteristics which cannot be changed without affecting the system design criteria. Coordination drawings may be categorized as mechanical, electrical interconnections, configuration and installation, operational sequence requirements, system switching, etc., as necessary. The notation "COORDINATION DRAWING" shall be shown, either adjacent to or as a subtitle in the title block.

3.8 Correlation drawing.

3.8.1 Definition. A correlation drawing discloses the physical and functional interface engineering requirements between or among components within a subsystem. Correlation drawings are used to correlate interface engineering data between a subsystem design activity and its subcontractors and vendors involved in the design of the subsystem.

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3.8.2 Requirements. The requirements prescribed for coordination drawings (3.7.2) shall apply also to correlation drawings, except that the delineation provided on correlation drawings shall be restricted to components within the subsystems. Correlation drawings shall contain no requirements intended to control the interfaces of associated subsystems. Should conflict exist between coordination and correlation drawings, data disclosed on the coordination drawings shall prevail. The notation "CORRELATION DRAWING" shall be shown, either adjacent to, or as a subtitle of the title block. Data prescribed on correlation drawings which cannot be changed without affecting cofunctioning subsystems shall be suitably identified.

3.9 Detail drawing.

3.9.1 Definition. A detail drawing discloses complete engineering requirements for the part(s) delineated thereon. A detail drawing also is used to delineate parts of buildings or structures, and to depict the relationship of the various parts, their sizes, contours and materials of construction (see figure 9).

3.9.1.1 Mono detail drawing. A mono detail drawing delineates a single part.

3.9.1.2 Multi detail drawing. A multi detail drawing delineates two or more parts (see figure 10).

3.9.2 Requirements. A detail drawing shall completely define all features of the part(s) depicted, including, as applicable, configuration, dimensions tolerances, materials, mandatory processes, surface finish, protective coating, symbols, etc. Documents required to supplement the drawing in stating end product requirements for the part(s) shall be prescribed by notes or tables on the drawing.

3.10 Detail-assembly drawing.

3.10.1 Definition. A detail-assembly drawing discloses an assembly on which one or more parts are detailed in the assembly view or on detail views, no separate detail drawings being prepared for such parts (see figure 11).

3.10.2 Requirements. Details of parts and assembly views on detail-assembly drawings shall have the completeness required by paragraphs 3.9.2 and 3.3.2, respectively. Parts detailed on a detail-assembly drawing and the assembly itself shall be assigned separate identifying numbers.

3.11 Diagram drawing.

3.11.1 Definition. A diagram drawing delineates the features and relationship of items forming an assembly or system by means of symbols and lines. A diagram drawing is a graphic explanation of the manner by which an installation, assembly or system (e.g., mechanical, electrical, electronic, hydraulic, pneumatic) performs its intended function.

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3.11.1.1 Electrical and electronic diagrams.

(a) Single line or one line diagram. A single line or one line diagram shows, by means of single lines and graphic symbols, the course of an electric circuit, or system of circuits and the component devices or parts used therein (see figure 12).

(b) Schematic or elementary diagram. A schematic or elementary diagram shows, by means of graphical symbols, the electrical connections and functions of a specific circuit arrangement. A schematic diagram facilitates tracing the circuit and its functions without regard to the actual physical size, shape, or location of the component devices or parts (see figure 13).

(c) Connection or wiring diagram. A connection or wiring diagram shows the electrical connections of an installation or of its component devices or parts. It may cover internal or external connections, or both, and contains such detail as is needed to make or trace connections that are involved. A connection diagram usually shows general physical arrangement of the component devices or parts (see figure 14).

(d) Interconnection diagram. An interconnection diagram is a form of connection or wiring diagram which shows only external connections between unit assemblies or equipments. See paragraph 3.4.3 concerning a form of inter-connection diagram known as a "running list" (see figure 15).

3.11.2 Requirements. Electrical and electronic diagrams shall disclose the requirements prescribed by American Standard ASA Y14.15, Electrical Diagrams.

3.11.3 Mechanical schematic diagram.

3.11.3.1 Definition. A mechanical schematic diagram illustrates the operational sequence or arrangement of a mechanical device (see figure 16).

3.11.3.2 Requirements. Sufficient detail shall be shown to explain the operational sequence or arrangement. The schematic may be isometric, semi-isometric, in perspective, or drawn as if all features were in a single plane. Nomenclature, symbols, part identifying numbers and dimensions should be shown as required.

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3.11.4 Piping diagram.

3.11.4.1 Definition. A piping (hydraulic, pneumatic or fluid) diagram depicts the interconnection of components by piping, tubing or hose, and when desired, the sequential flow of fluids in the system (see figure 17).

3.11.4.2 Requirements. Sufficient detail shall be shown to explain (a) the arrangement of the piping, valves, etc., or (b) the operational sequence. Symbolic line representation may be used to distinguish the functions of the various parts. When the objective is to show the arrangement, characteristics such as the following may be shown: routing of fluids, physical locations and arrangement of components, pipe diameters, types and sizes of fittings, flow, pressure, volume, etc.

3.12 Elevation drawing.

3.12.1 Definition. An elevation drawing depicts vertical projections of buildings or structures; or profiles of equipment such as aircraft, automotive and marine, or portions of the same (see figure 18).

3.12.2 Requirements. As applicable, an elevation drawing shows configuration, shapes and sizes of features, walls, bulkheads, compartments, assignment of space, location and arrangement of machinery or fixed equipment. An elevation drawing may indicate materials of construction.

3.13 Envelope drawing.

3.13.1 Definition. An envelope drawing discloses the physical and functional engineering requirements for a vendor designed item, such requirements being delineated to the extent necessary to enable the development of the design details. Distinctive from control drawings (3.6), which apply to commercial and vendor items usually existent, envelope drawings apply to new items, vendor designed to suit a specific design application.

3.13.2 Requirements. An envelope drawing shall disclose, as applicable, configuration, dimensions of envelope (expressed as limits or as maxima), mounting and mating dimensional limits, and all other interface characteristics that affect the physical interchangeability of the item. Additionally, performance, reliability, maintainability, environmental and other requirements, as necessary, to meet design requirements, and acceptance test requirements to the extent necessary to ensure that such design requirements have been met, shall also be specified, either on the drawing or in a document invoked thereon. Information to establish external connections, if any, shall be included on the drawing, or in a referenced document.

3.14 Erection drawing.

3.14.1 Definition. An erection drawing shows procedures and the operation sequence for erection or assembly of individual items or assemblies of items.

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3.14.2 Requirements. An erection drawing shall show the location of each part in the structure, the identification markings, the types of fastenings required, the approximate weight of heavy structural members, the controlling dimensions, and any other information which will contribute to the expeditious erection of the structure.

3.15 Inseparable assenbly drawing.

3.15.1 Definition. An inseparable assembly drawing delineates items (pieces) which are separately fabricated and are permanently joined together (e.g. welded, brazed, etc.) to form an integral unit (part). (see figure 19).

3.15.2 Requirements. An inseparable assembly drawing shall fully define the end product as assembled. The pieces of the inseparable assembly may be detailed either on separate detail drawings or on the inseparable assembly drawing itself; however, all dimensions and characteristics of the pieces that are not also characteristics of the end product are informational, rather than mandatory, and should be labeled as reference information, or "INFO".

3.16 Installation drawing.

3.16.1 Definition. An installation drawing shows general configuration, attaching hardware and complete interface information necessary to locate, position, mount and attach an item relative to its supporting structure, or to other items (see figure 20).

3.16.2 Requirements. An installation drawing shall show all mounting information, including mounting hole or bolt sizes, mounting hole locations, alignment, and other mounting requirements. When applicable, piping and cabling data shall be included, unless separate drawings of interconnecting data are furnished. Exterior views and dimensions in plan and elevation shall be shown, including necessary clearances, unless clearance information is shown on a separate installation control drawing.

3.17 Installation control drawing.

3.17.1 Definition. An installation control drawing sets forth dimensional information for an item in terms of area and space, access clearances, training clearances, pipe and cable attachments required for the installation and cofunctioning of the item to be installed with related items. Installation control drawings are sometimes called "space control" or "working circle" drawings (see figure 21).

3.17.2 Requirements. An installation control drawing shall include the following information, as applicable:

(a) Overall and principal dimensions in sufficient detail to establish the limits of space in all directions required for installation,

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operation and servicing. The amount of clearance required to permit the opening of doors or the removal of plug-in units shall be included. Clearance for travel or rotation of any moving parts shall be shown, including the centers of rotation, angles of train in azimuth, elevation and depression, and radii from each pivot to the end of each rotating element involved in clearance determinations.

When not shown on a separate installation drawing:

(b) Information necessary for preparation of foundation plans, including mounting plate details, drilling plans and shock mounting details.

(c) Location, size and arrangement of any ventilation ducts supplied or required.

(d) Weight of each unit which is separately mounted.

(e) Location, type and dimensions of cable entrances, terminal tubes and electrical connectors.

(f) When not shown on separate drawings, interconnecting and cabling data.

3.18 Kit drawing.

3.18.1 Definition. A kit drawing indicates or depicts a packaged unit consisting of an item or group of items, instructions, photographs or drawings such as are used in modification, installation or conversion. The items in a kit normally do not in themselves constitute a complete functional assembly. A kit drawing may be a listing of part numbers, a pictorial representation of parts or a combination of both (see figure 22).

3.18.2 Requirements. A kit drawing shall include or reference a list of the entire kit contents, including part numbers and part nomenclature and quantities required.

3.19 Master plan drawing.

3.19.1 Definition. A master plan drawing is used in the architectural, topographical or construction field to show sufficient features that it may be used as a guide in long range development of areas.

3.19.2 Requirements. A master plan drawing shall show, as necessary, section boundary lines, horizontal and vertical control datums, acreage, location and description of each existing and proposed structure, existing and proposed surfaced and unsurfaced roads and sidewalks, streams, rights of way, and any appurtenances required; existing and proposed utilities, sewer and water lines, north point indicator, contour lines and profiles.

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3.20 Matched parts drawing.

3.20.1 Definition. A matched parts drawing depicts parts which are machine matched, or otherwise mated, and for which replacement as a matched set or pair is essential (see figure 23).

3.20.2 Requirements. The operating or mating characteristics of the matched set shall be stated. The matched parts may be detailed on the matched parts drawing or on separate drawings. A single part number shall be assigned to each of the matched parts, the drawing shall require identification marking of the matched set. "Furnish only as a matched set" or similar note should be on the drawing.

3.21 Multi-sheet drawing.

3.21.1 Definition. A multi-sheet drawing consists of two or more sheets applicable to the same item. A multi-sheet drawing is not a type of drawing in the same sense as a detail or assembly drawing; rather, the term "multi-sheet drawing" describes the use of several sheets in presenting all of the information applicable to a single drawing of any one of several types, as opposed to the use of a single roll or flat sheet.

3.22 Optical elements and optical systems drawings.

3.22.1 Definitions and requirements. Definitions and requirements for drawings pertaining to optical elements and optical systems are stated in Standard MIL-STD-34.

3.23 Plan drawing.

3.23.1 Definition. A plan drawing depicts a horizontal projection of a structure, showing the layout of the foundation, floor, deck, roof, or utility system (see figure 24).

3.23.2 Requirements. As applicable, a plan drawing shall show shapes, sizes and materials of the foundation, its relation to the superstructure and its elevation with reference to a fixed datum plane, location of walls, partitions, bulkheads, stanchions, companion ways, openings, columns, stairs, shapes and sizes of roofs, parapet walls, drainage, skylights, ventilators, etc. A plan drawing shall specify materials of construction and shall show the arrangement of structural framing. As applicable, the location of equipment or funiture may be indicated. Also a plan drawing for services may depict individual layouts for heating, plumbing, air conditioning, electrical or other utility systems.

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3.24 Plot (plat) plan drawing.

3.24.1 Definition. A plot (or plat) plan drawing represents areas on which structures are clearly indicated with detailed information regarding their relationship to other structures, existing and proposed utilities, topography, boundary lines, roads, walks, fences, etc.

3.24.2 Requirements. A plot (plat) plan drawing shall show, as necessary, the property lines and locations, contours and profiles, shrubbery, existing and new utilities, sewer and water lines, building lines, location of structures to be constructed, existing structures, approaches, finished grades, and other pertinent data.

3.25 Tabulated drawing.

3.25.1 Definition. A tabulated drawing depicts similar items, which as a group, have some constant and some variable characteristics. A tabulation drawing precludes the preparation of an individual drawing for each item (see figure 25).

3.25.2 Requirements. The difference (variables) between the items defined by a tabulated drawing shall be tabulated, and the fixed (constant) characteristics shall be depicted or stated only once. Normally, a pictorial representation of a single item is shown, with the variable dimensions coded by means of letters, which are used as headings for columns in the tabulation. The variables then are entered in the table under the appropriate headings and on the same line as the identifying number or letter of the item to which they pertain. The statement of requirements shall be as complete as that required by paragraph 3.9.2 for a single part.

3.26 Tube bend drawing.

3.26.1 Definition. A tube bend drawing discloses, by means of pictorial or tabular delineation, or combinations thereof, complete bend data required for the fabrication of a multi-plane rigid metal tube. Two basic methods of delineating tube bend data are illustrated (see figures 26 and 27).

3.26.2 Requirements. A tube bend drawing shall disclose, as applicable, tube material, type of ends, identification, quantity and designation of fittings, bend radii and angles, intersection points, intermediate and overall lengths and all other data required for fabrication of the item. Tubing of complex configuration shall preferably be disclosed by utilizing combinations of pictorial and tabular delineation; while tubing having simple configuration may be defined by means of tabular presentation.

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3.27 Undimensioned drawing.

3.27.1 Definition. An undimensioned drawing depicts, to a precise scale, usually on environmentally stable material, loft line information, a template, a pattern, or a printed circuit.

3.27.2 Requirements.

3.27.2.1 Printed wiring. The requirements for master drawings pertaining to printed wiring are stated in Standard MIL-STD-275.

3.27.2.2 Undimensioned drawings other than printed wiring. Undimensioned drawings shall include all information necessary to fabricate parts within the established manufacturing limits set forth in other engineering data.

3.28 Vicinity plan drawing.

3.28.1 Definition. A vicinity plan drawing (or vicinity map used with construction drawings) delineates the relationship of a site to features of the surrounding area, such as towns, bodies of water, railroads, highways, etc.

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Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Copies of this standard for military use may be obtained as indicated in the foreword to, or the general provisions of, the Index of Military Specifications and Standards.

Copies of this standard may be obtained for other than official use by individuals, firms, and contractors from the Superintendent of Documents, U. S. Government Print Officer, Washington, D. C. 20402

Both the title and identification symbol should be stipulated when requesting copies of military standards.

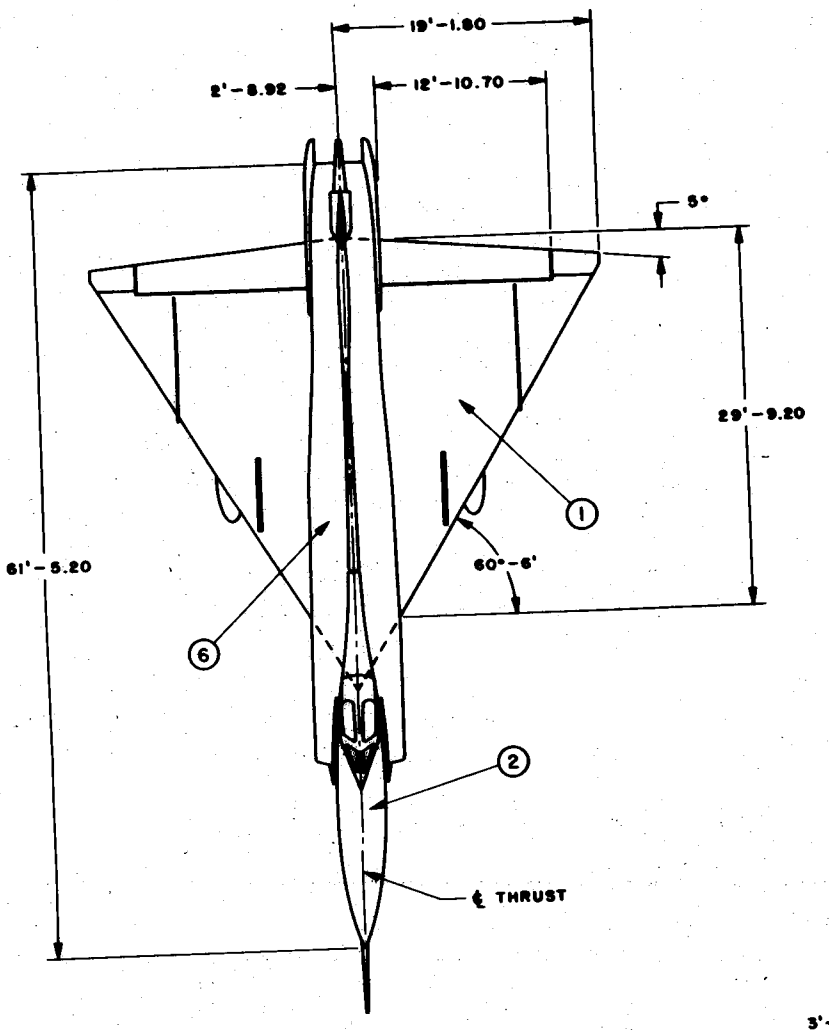
Custodians:

Army - MO
Navy - WEP
Air Force - AFLC

Preparing activity:

Navy - WEP
(Project DRPR-0033)

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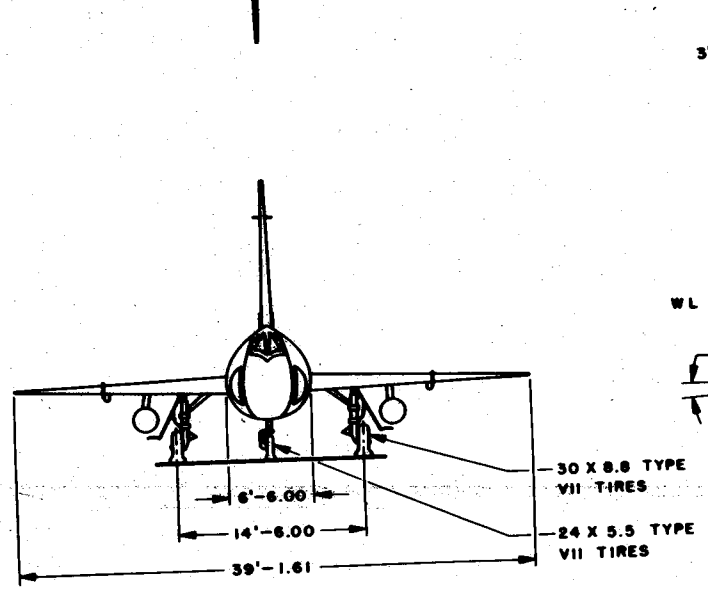
DATA

FULL LOAD GROSS WEIGHT	LB	22462
WEIGHT EMPTY	LB	12620
USEFUL LOAD	LB	9842
ALTERNATE GROSS WEIGHT	LB (FERRY)	25400
ALTERNATE USEFUL LOAD	LB (FERRY)	12780
WING LOADING	LB / FT	150
THRUST LOADING	LB / LB THRUST T.O.	1.40
ENGINE MODEL DESIGNATION	P & W J57-P23 (16000 LB THRUST)	
WING AREA	FT²	695.05
ASPECT RATIO		2.189
AIRFOIL	NACA .0004-65 (MOD)	
DIHEDRAL	DEG	0
INCIDENCE	DEG	0
ELEVON'S AREA TOTAL	FT²	67.15
DEFLECTION	DEG	34° UP, 19° DN
VERTICAL TAIL AREA	FT²	95.14
ASPECT RATIO		1.38
AIRFOIL	NACA .0004-65 (MOD)	
RUDDER AREA	FT²	9.70
DEFLECTION	DEG	±30°
MANUFACTURER'S NO.	MODEL 8-10	

CONTROL SURFACES DATA

	AILERON	ELEVATOR	TOTAL
UP	7°	25°	32°
DOWN	7°	10°	17°

ITEM NO.	NOMENCLATURE	DWG NO.
1	WING	0000020
2	FUSELAGE	0000021
3	VERTICAL STABILIZER	0000022
4	MAIN LANDING GEAR	0000023
5	NOSE WHEEL LANDING GEAR	0000024
6	POWER PLANT INSTALLATION	0000025



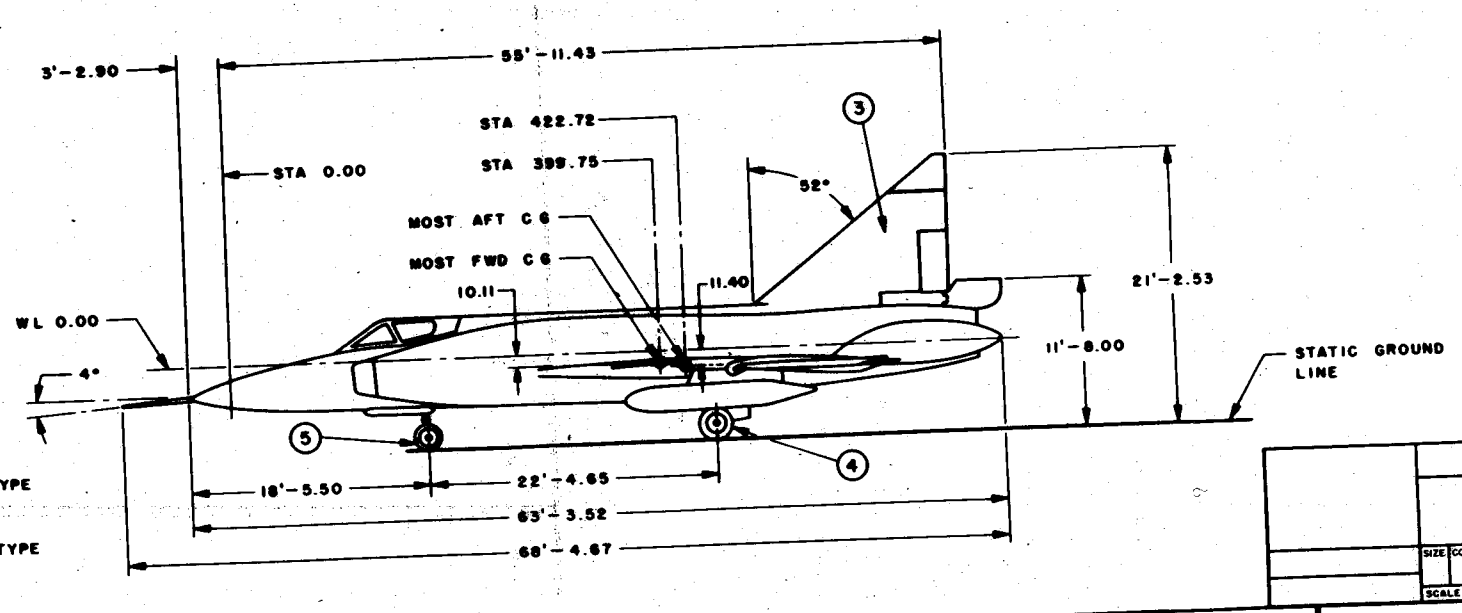


FIGURE 1. Sample Arrangement Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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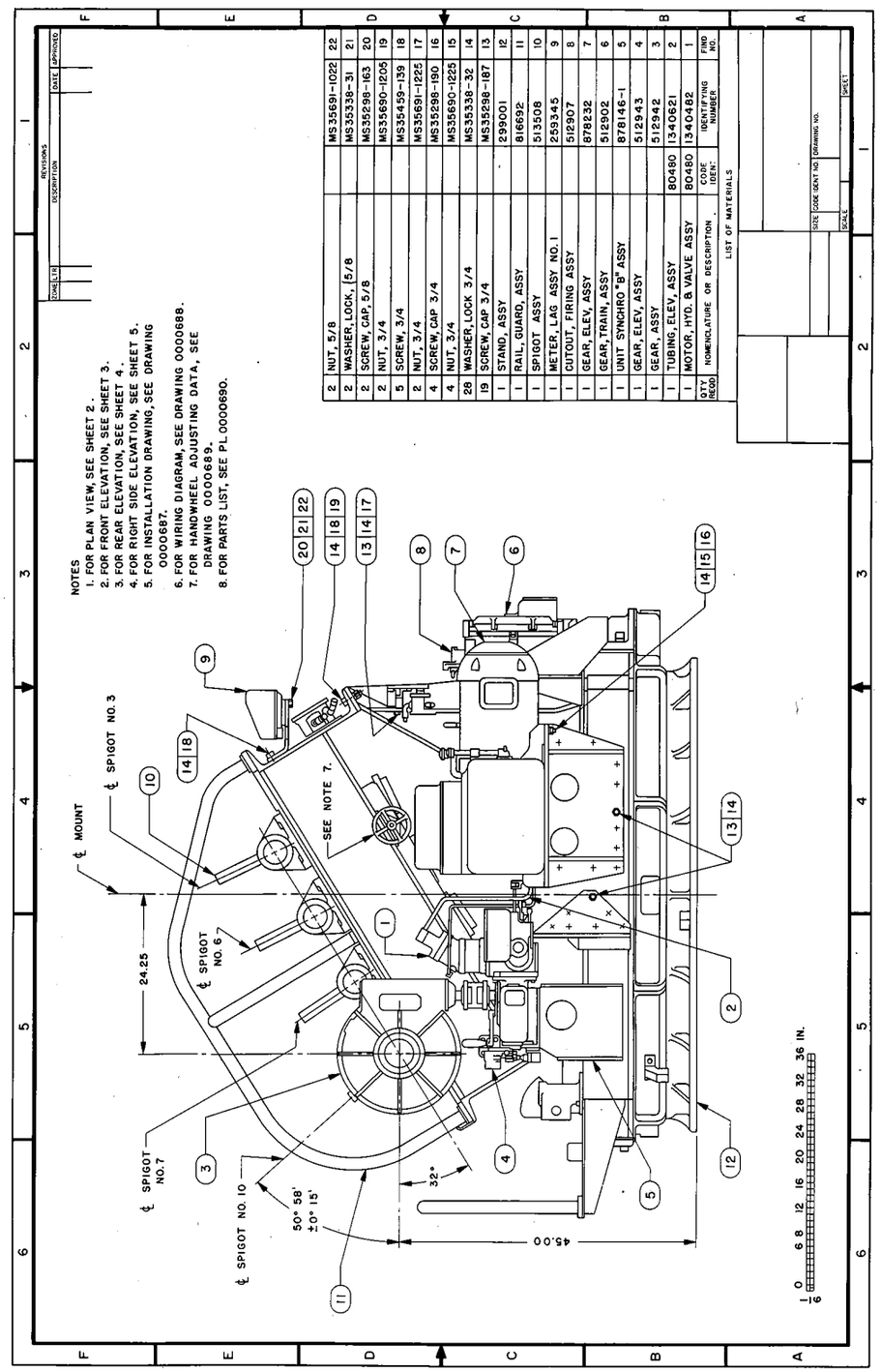


FIGURE 2. Sample Assembly Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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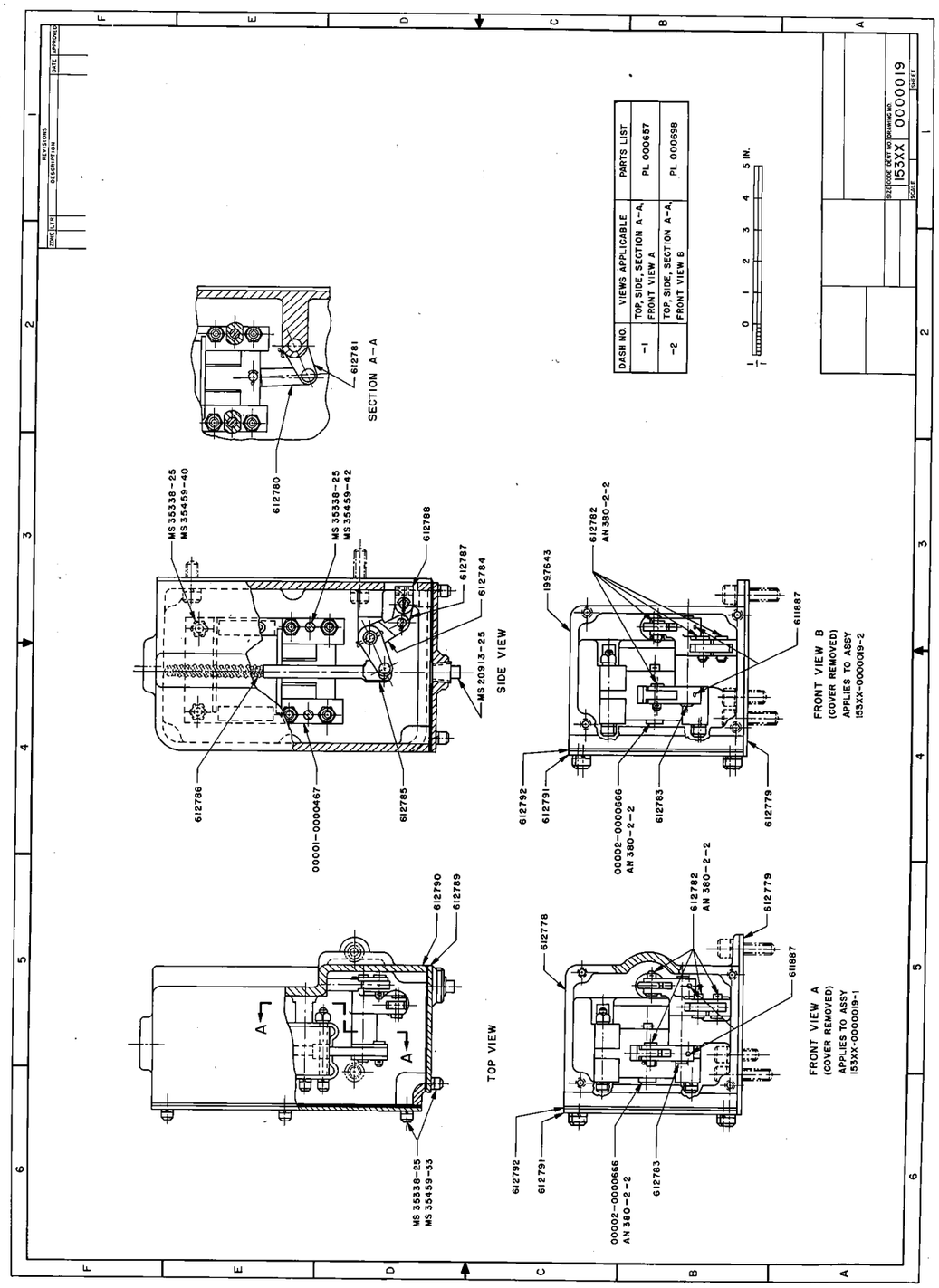


FIGURE 3. Sample Assembly Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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NOTES:

1. ALL CONDUCTOR ENDS TO BE STRIPPED .50 IN AND TINNED BEFORE ASSEMBLY.
2. USE QQ-S-571 SN 60 SOLDER FOR ALL SOLDERED CONNECTIONS.
3. INSTALL CRIMP STYLE TERMINALS IN ACCORDANCE WITH MIL-T-00000.
4. BIND WIRES TOGETHER USING NYLON CORD, MIL-C-572, FORM S OR Y LOCKSTITCH AT .50 INCH SPACING.
5. TERMINATE LACING AT SINGLE CONDUCTOR BRANCH-OFFS; AT .75 IN. (MIN) FROM CONDUCTOR ENDS ON MULTIPLE CONDUCTOR BRANCH-OFFS; OR AS INDICATED ON DRAWING.
6. ALL DIMENSIONS TO HAVE A TOLERANCE OF ±.12 TOLERANCE NOT ACCUMULATIVE.
7. SLEEVING SHALL BE REQUIRED WHEREVER CONDUCTORS ARE NOT DIRECTLY CONNECTED TO TERMINAL BLOCKS (CRIMP TERMINALS, JACKS, PLUGS) OR WHEREVER HARNESS SHALL BE EXPOSED TO MECHANICAL WEAR.

WIRE NO.	COLOR BODY	COLOR FIRST TRACER	COLOR SECOND TRACER	APPROX. LENGTH REQUIRED FROM	TO	TOTAL (INCHES)	REFERENCE
1	BLK	-	-	J101-1	TB101-3		MIL-W-16878 TYPE C-20
2	"	-	-	J103-E	TB101-3		"
3	"	-	-	J102-2	T102-2		"
4	BLK	-	-	TB102-9	TB101-3		MIL-W-16878 TYPE C-20
11	BRN	-	-			44	
12	"	-	-	XV102-2	XV101-2		MIL-W-16878 TYPE C-20
13	"	-	-	XV102-2	XV103-2		"
14	BRN	-	-	XV103-2	T101-2		MIL-W-16878 TYPE C-20
18	RED	-	-	XV103-2	XV104-2		"
19	"	-	-	TB102-3	TB101-11	37	MIL-W-16878 TYPE C-24
20	"	-	-	TB101-11	TB103-1		"
21	RED	-	-	TB103-1	C108-A		MIL-W-16878 TYPE C-24
25	ORN	-	-	TB103-4	C108-B		"
26	"	-	-	TB101-10	XV101-6	40	MIL-W-16878 TYPE C-20
27	ORN	-	-	TB103-4	XV104-4		"
31	YEL	-	-	XV103-4	XV104-4		MIL-W-16878 TYPE C-20
32	YEL	-	-	TB102-8	XV102-8	38	MIL-W-16878 TYPE C-20
36	ORN	-	-	TB101-6	XV101-5		MIL-W-16878 TYPE C-20
37	"	-	-	J101-2	TB101-1	26	MIL-W-16878 TYPE C-20
38	"	-	-	J103-C	XV102-5		"
39	"	-	-	J103-D	TB101-13		"
40	"	-	-	TB102-6	XV102-5		"
41	GRN	-	-	TB102-14	TB101-13		"
46	BLU	-	-	TB101-4	XV101-4		MIL-W-16878 TYPE C-20
47	BLU	-	-	TB102-4	XV102-3	62	MIL-W-16878 TYPE C-20
52	GY	-	-	TB101-14	XV101-8		MIL-W-16878 TYPE C-20
53	"	-	-	J104-B	T101-B	27	MIL-W-16878 TYPE C-20
54	GY	-	-	J104-A	J103-F		"
57	WHT	-	-	J103-A	T101-A		MIL-W-16878 TYPE C-20
60	WHT	BLK	BRN	J103-B	TB102-13	60	MIL-W-16878 TYPE C-20
61	WHT	BLK	BRN	XV105-8	T101-5	12	MIL-W-16878 TYPE C-18
64	WHT	BLK	RED	XV105-2	T101-6		MIL-W-16878 TYPE C-18
65	"	"	"	TB103-3	C108-C	20	MIL-W-16878 TYPE C-18
66	WHT	BLK	RED	T101-6	T102-4		"
69	WHT	BLK	ORN	T101-6	C108-C		MIL-W-16878 TYPE C-18
70	WHT	BLK	ORN	TB102-1	XV103-5	23	MIL-W-16878 TYPE C-18
73	WHT	BLK	GRN	TB102-11	XV104-5		MIL-W-16878 TYPE C-18
74	WHT	BLK	GRN	XV103-3	T102-5	41	MIL-W-16878 TYPE C-18
77	WHT	BLK	BLU	XV104-3	T102-3		MIL-W-16878 TYPE C-18
78	WHT	BLK	BLU	XV105-4	T101-3	14	MIL-W-16878 TYPE C-18
81	WHT	BRN	RED	XV105-6	T101-4		MIL-W-16878 TYPE C-18
				J102-1	T102-1	25	MIL-W-16878 TYPE C-18
						20	MIL-W-16878 TYPE C-18

REFERENCE DESIGNATION	NOMENCLATURE	IDENTIFICATION	AMOUNT REQUIRED
	INSULATION SLEEVING, ELECTRICAL	MIL-I-631, TYPE F, FORM U, CLASS 1, .034 ID	4 INCHES
	INSULATION SLEEVING, ELECTRICAL	MIL-I-631, TYPE F, FORM U, CLASS 1, .076 ID	3 INCHES
	INSULATION SLEEVING, ELECTRICAL	MIL-I-631, TYPE F, FORM U, CLASS 1, .095 ID	9 INCHES
J103	CONNECTOR, RECP, 6 PIN	MS 00000-00-000	1

REFERENCES:

1. LIST OF MATERIAL LM 0000810
2. SCHEMATIC DIAGRAM 0000812
3. WIRING DIAGRAM 0000811
4. ASSEMBLY DRAWING 0000813

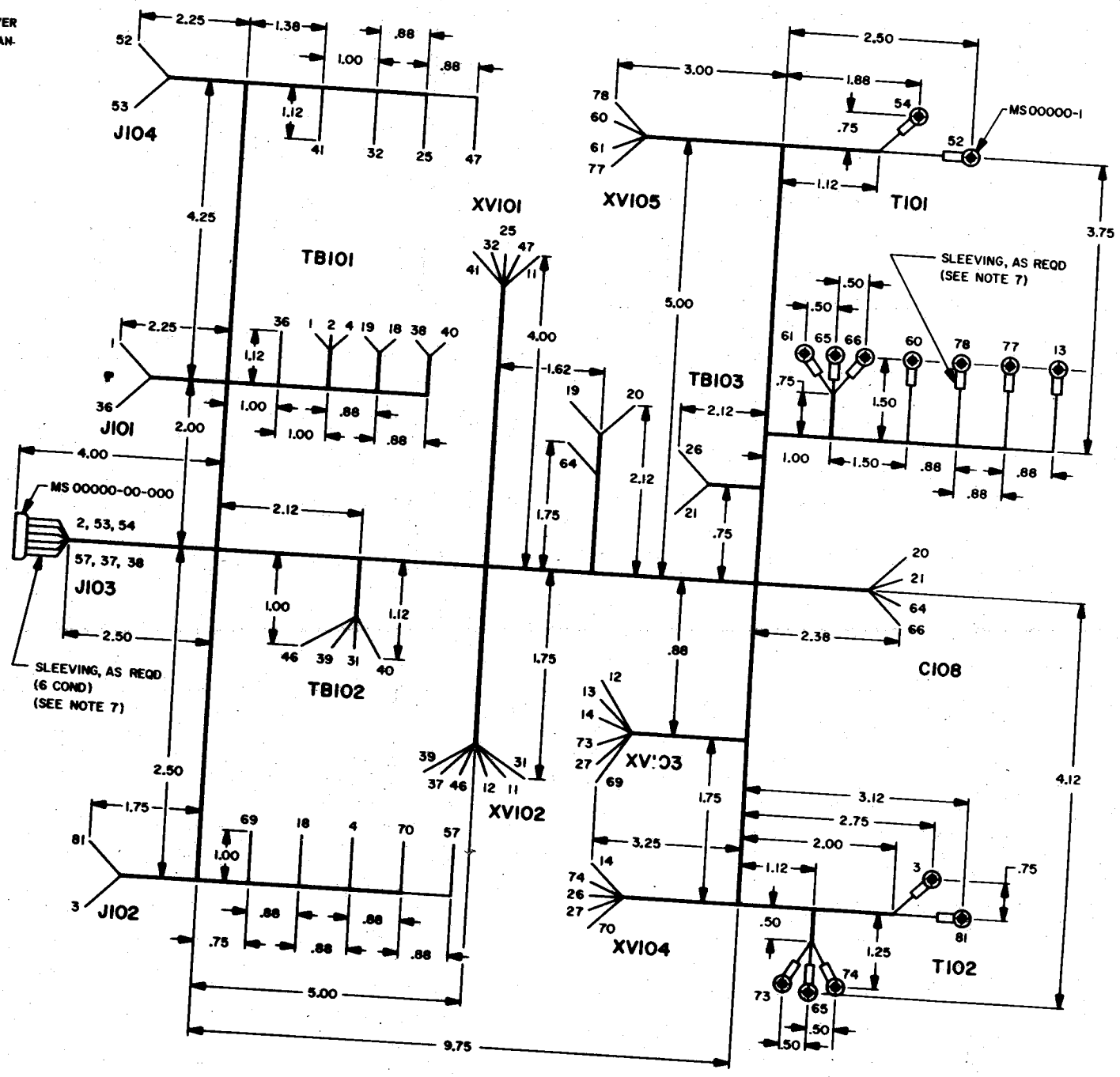
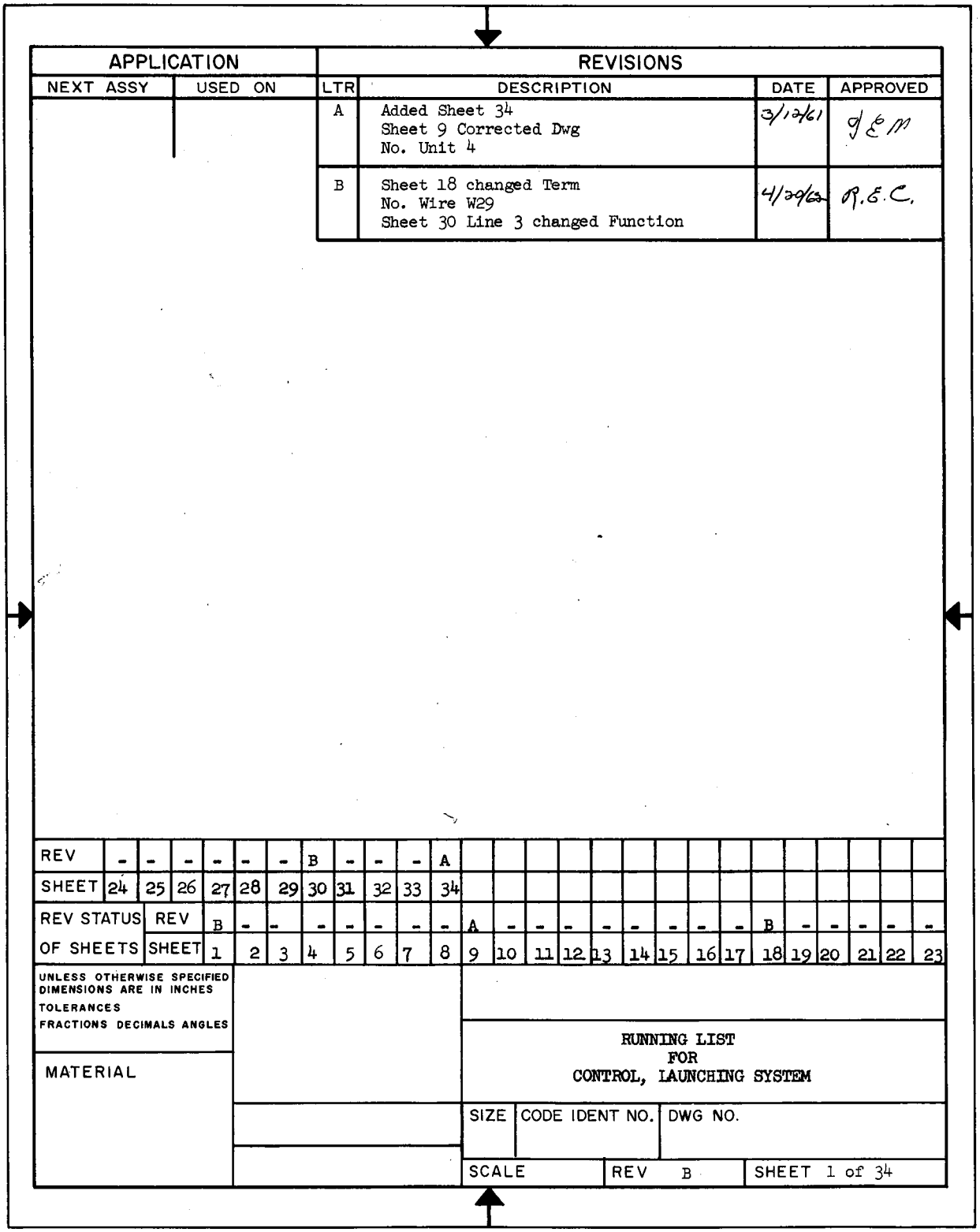


FIGURE 4. Sample Wiring Harness Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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APPLICATION		REVISIONS			
NEXT ASSY	USED ON	LTR	DESCRIPTION	DATE	APPROVED
		A	Added Sheet 34 Sheet 9 Corrected Dwg No. Unit 4	3/12/61	J&M
		B	Sheet 18 changed Term No. Wire W29 Sheet 30 Line 3 changed Function	4/29/62	R.E.C.

REV	-	-	-	-	-	-	B	-	-	-	A														
SHEET	24	25	26	27	28	29	30	31	32	33	34														
REV STATUS	REV	B	-	-	-	-	-	-	-	A	-	-	-	-	-	-	-	-	B	-	-	-	-	-	
OF SHEETS	SHEET	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES FRACTIONS DECIMALS ANGLES		
MATERIAL		RUNNING LIST FOR CONTROL, LAUNCHING SYSTEM
		SIZE / CODE IDENT NO. / DWG NO.
		SCALE / REV B / SHEET 1 of 34

FIGURE 5(a). Sample Book Form Drawing. Title Sheet

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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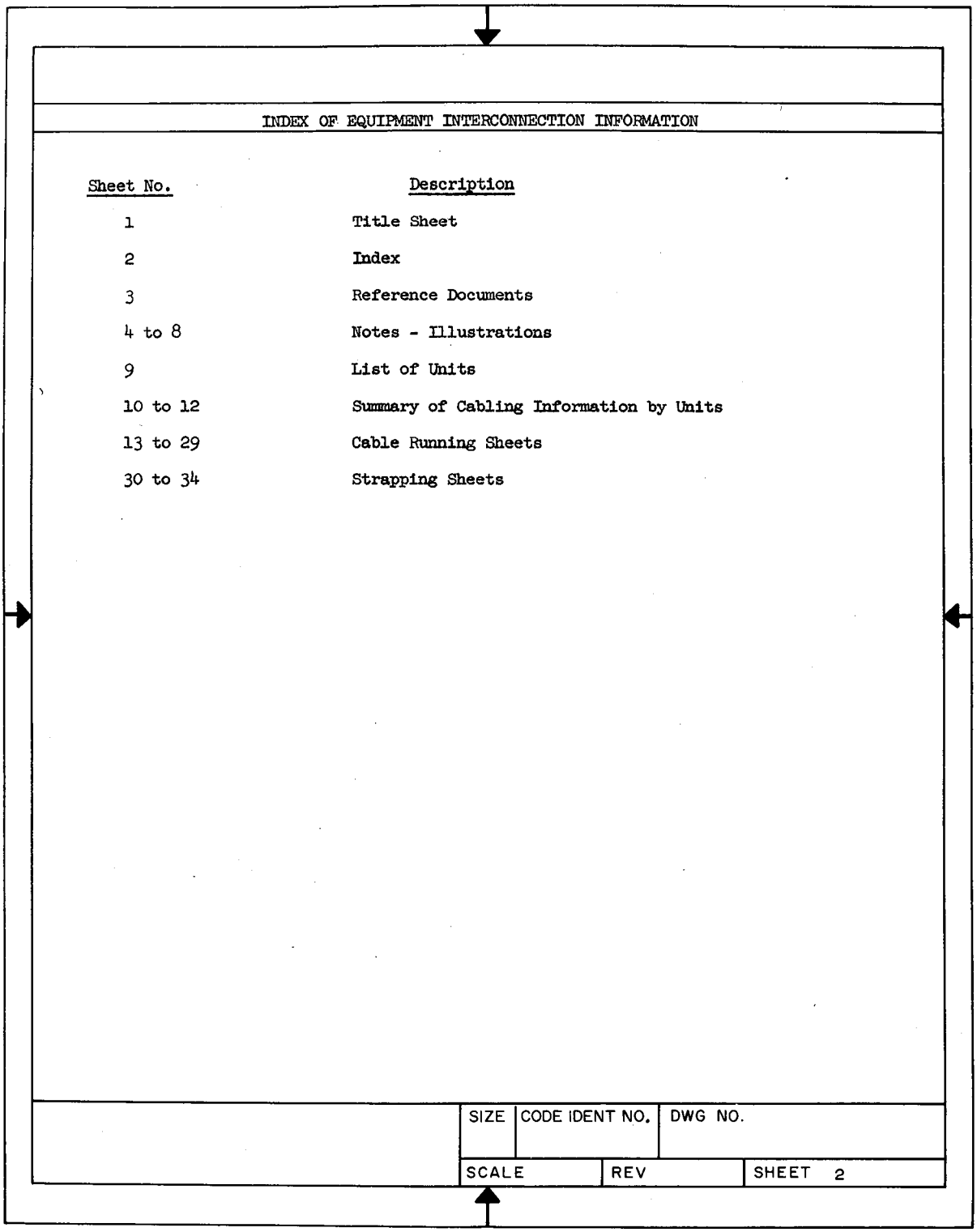


FIGURE 5(b). Sample Book Form Drawing. Index of Equipment Interconnection Information

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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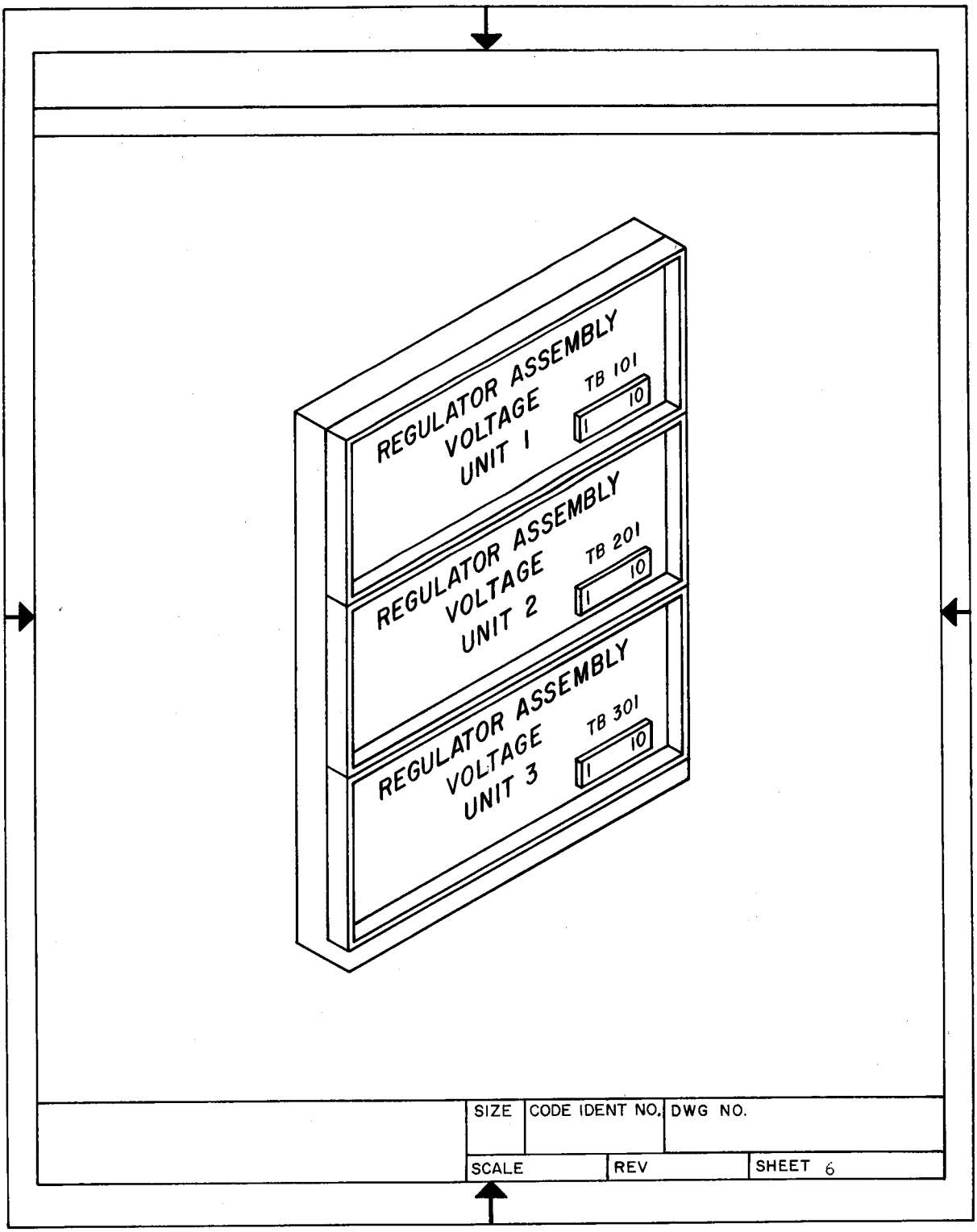
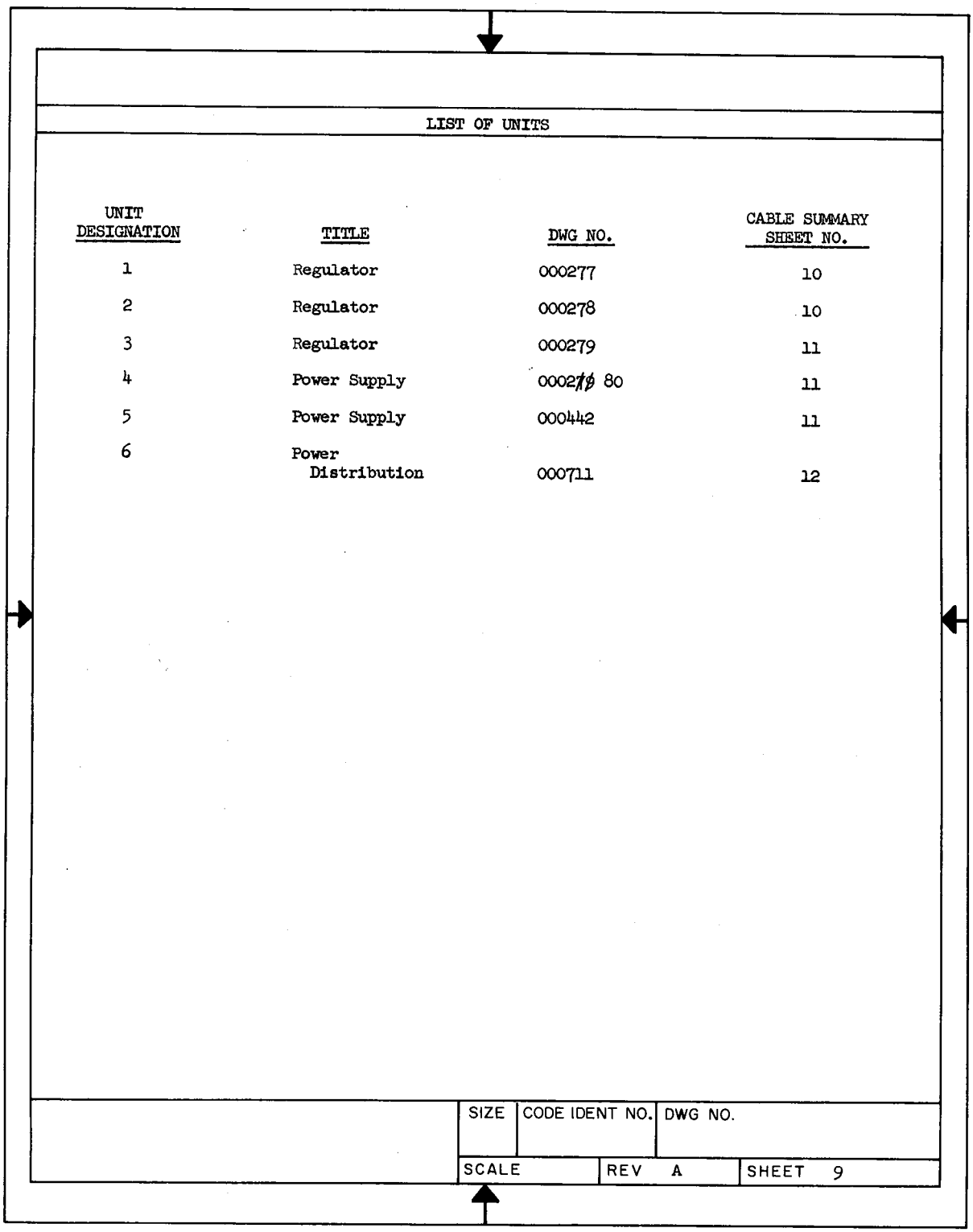


FIGURE 5(c). Sample Book Form Drawing. Regulator

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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LIST OF UNITS

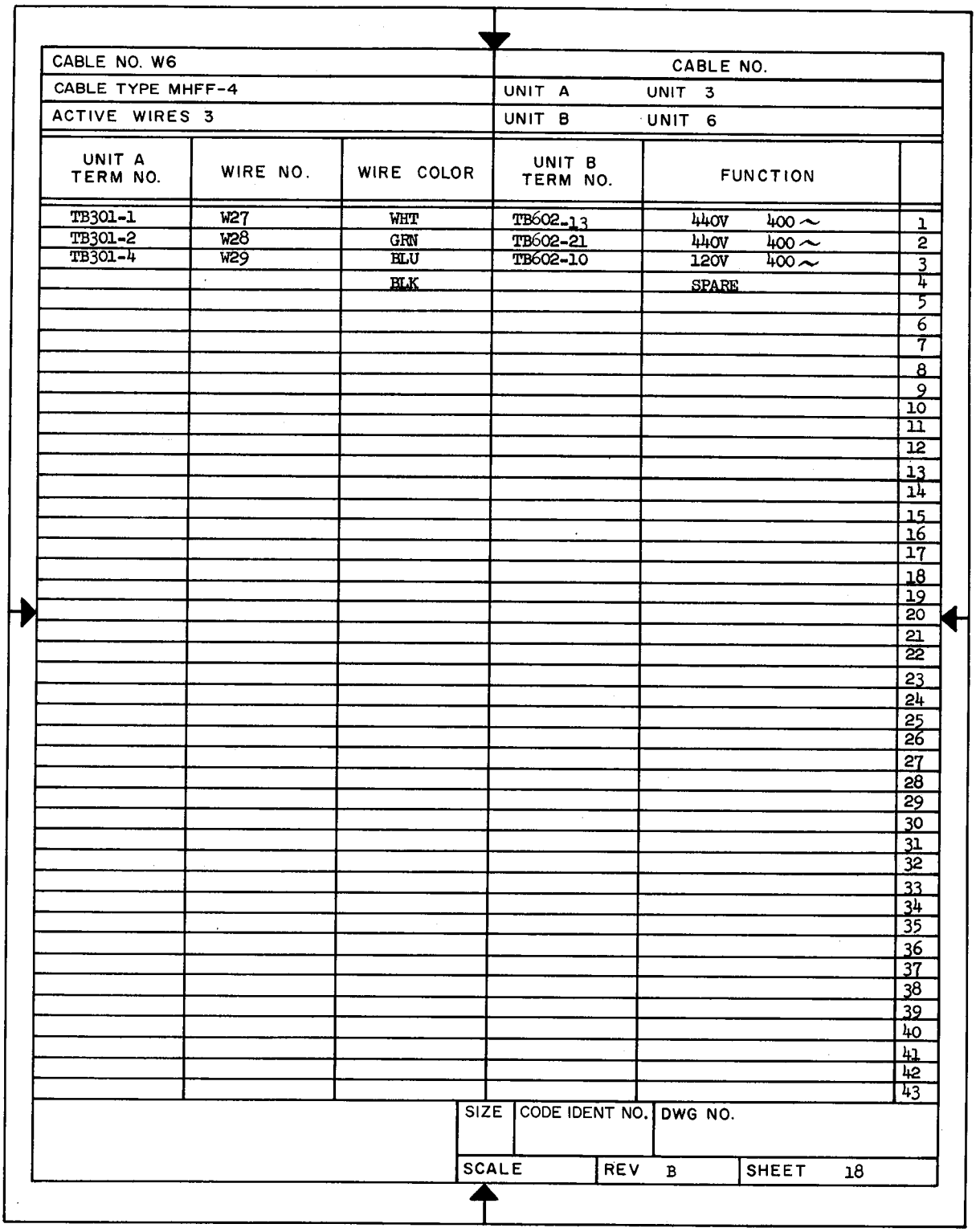
UNIT DESIGNATION	TITLE	DWG NO.	CABLE SUMMARY SHEET NO.
1	Regulator	000277	10
2	Regulator	000278	10
3	Regulator	000279	11
4	Power Supply	000280	11
5	Power Supply	000442	11
6	Power Distribution	000711	12

	SIZE	CODE IDENT NO.	DWG NO.
	SCALE	REV A	SHEET 9

FIGURE 5(d). Sample Book Form Drawing. List of Units

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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CABLE NO. W6			CABLE NO.		
CABLE TYPE MHFF-4			UNIT A UNIT 3		
ACTIVE WIRES 3			UNIT B UNIT 6		
UNIT A TERM NO.	WIRE NO.	WIRE COLOR	UNIT B TERM NO.	FUNCTION	
TB301-1	W27	WHT	TB602-13	440V 400 ~	1
TB301-2	W28	GRN	TB602-21	440V 400 ~	2
TB301-4	W29	BLU	TB602-10	120V 400 ~	3
		BLK		SPARE	4
					5
					6
					7
					8
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					42
					43

SIZE	CODE IDENT NO.	DWG NO.
SCALE	REV B	SHEET 18

FIGURE 5(e). Sample Book Form Drawing. Cable Running Sheet

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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NOTES:

1. INTERPRET DRAWING IN ACCORDANCE WITH STANDARDS PRESCRIBED BY MIL-D-70327.

2. GENERAL REQUIREMENTS:
   A. TRANSFORMER MUST MEET THE GENERAL REQUIREMENTS OF MIL-T-27 FOR GRADE 4, CLASS R, LIFE EXPECTANCY "X" TRANSFORMER, WITH RESPECT TO MATERIAL, WORKMANSHIP, GENERAL DESIGN FEATURES AND TEST REQUIREMENTS.

3. ELECTRICAL REQUIREMENTS:
   A. PRIMARY, 109, 115, AND 121 VOLTS RMS, 57 TO 63 CYCLES, SINGLE PHASE.
   B. SECONDARY, CENTERTAPPED AT TERMINAL 6.
   VOLTAGE: 25.0 VOLTS RMS LOADED.
   DIRECT CURRENT RESISTANCE: 2.0 OHMS MAXIMUM.
   LOAD CURRENT 1.7 AMPERES RMS
   WORKING VOLTAGE: CENTERTAP SHALL BE GROUNDED.
   DIELECTRIC TEST VOLTAGE: 500 VOLTS RMS.
   C. SHALL HAVE AN ELECTROSTATIC SHIELD BETWEEN PRIMARY AND SECONDARY. SHIELD, CORE, AND CORE BRACKETS SHALL BE ELECTRICALLY AND MECHANICALLY SECURED TO MOUNTING DEVICES.

4. ENVIRONMENTAL REQUIREMENTS:
   A. AMBIENT TEMPERATURE RANGE: MINUS 10 TO PLUS 65 DEGREES CENTIGRADE.
   B. MAXIMUM OPERATING TEMPERATURE: 105 DEGREES CENTIGRADE.

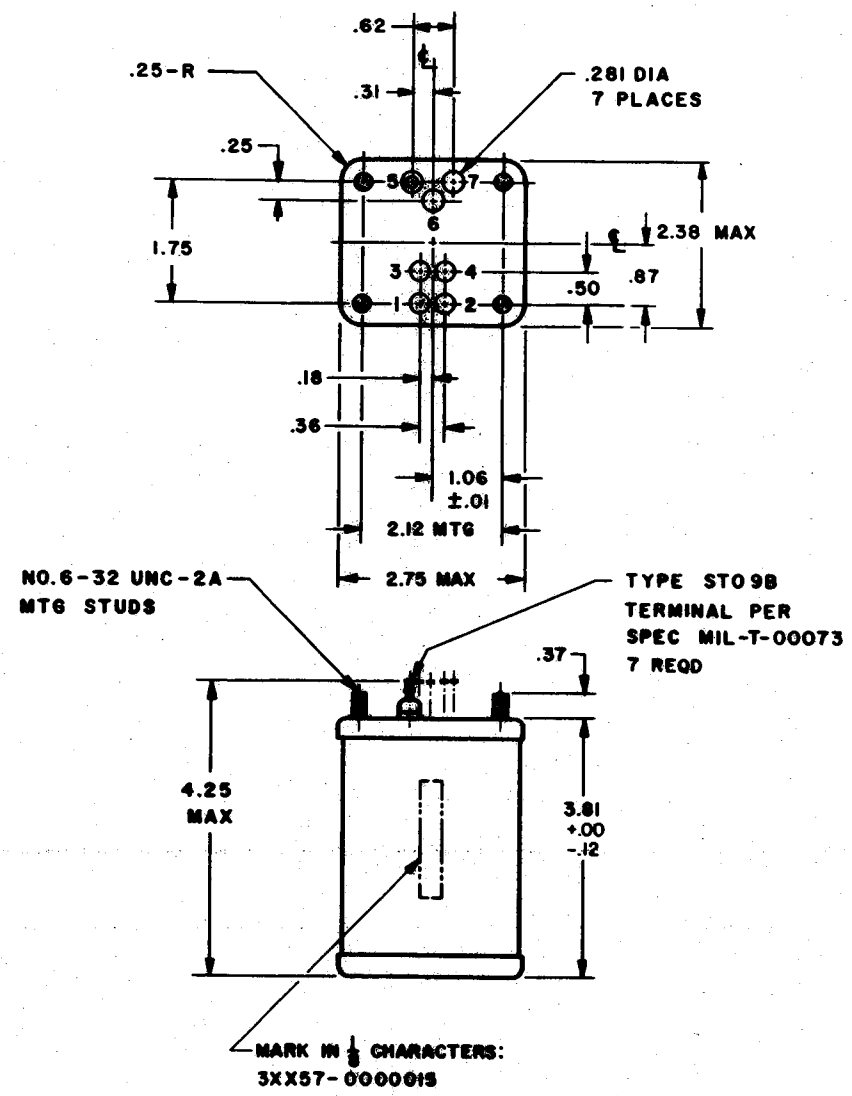
5. CONSTRUCTION REQUIREMENTS:
   A. HERMETICALLY SEALED, METAL CASED.
   B. FINISH: PRIME WITH ONE COAT ZINC CHROMATE, SPEC MIL-P-8585 AND FINISH WITH ONE COAT GRAY ENAMEL, SPEC MIL-E-15090. COLOR SHALL BE IN ACCORDANCE WITH FED STD NO. 595, COLOR CHIP 16376.
   C. MARKING: TERMINALS SHALL BE PERMANENTLY AND LEGIBLY MARKED IN ACCORDANCE WITH SCHEMATIC DIAGRAM.

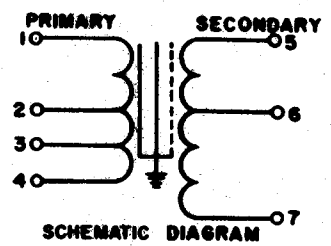
6. ONLY THE ITEM DESCRIBED ON THIS DRAWING WHEN PROCURED FROM THE VENDOR(S) LISTED HEREON IS APPROVED BY THE XYZ AEROSPACE COMPANY, LOS ANGELES 15, CALIFORNIA FOR USE IN THE APPLICATION(S) SPECIFIED HEREON. A SUBSTITUTE ITEM SHALL NOT BE USED WITHOUT PRIOR TESTING AND APPROVAL BY THE XYZ AEROSPACE COMPANY OR BY THE NAVAL AIR ENGINEERING CENTER, PHILADELPHIA 12, PA.

APPROVED SOURCE(S) OF SUPPLY		
VENDOR	VENDOR'S ITEM IDENT NO.	APPLICATION
JOHN SMITH, INC. MILWAUKEE, WIS.	6041H7	PROJECTOR MK. 5 MOD. 7

REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPROVED





SOURCE CONTROL DRAWING

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ON FRACTIONS DECIMALS ANGLES ±.02

MATERIAL:

SIZE	CODE IDENT NO.	DRAWING NO.
	3XX57	0000015

SCALE SHEET

FIGURE 6. Sample Source Control Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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NOTES:

1. INTERPRET DRAWING IN ACCORDANCE WITH STANDARDS PRESCRIBED BY MIL-D-70327.

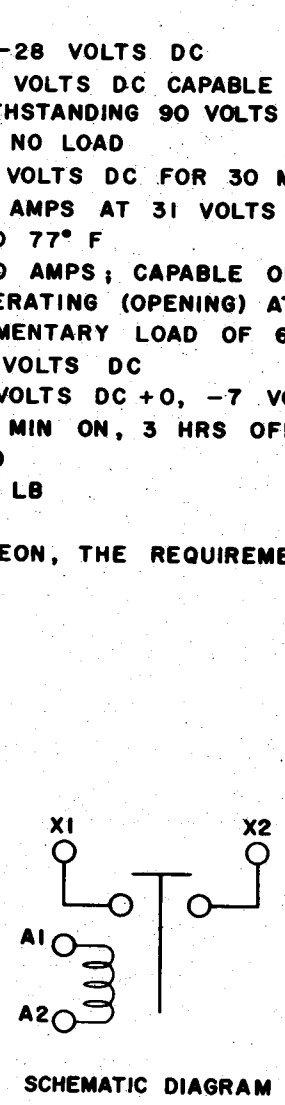
2. REQUIREMENTS
   - (A) NOMINAL OPERATING VOLTAGE
     - COIL _____ 24-28 VOLTS DC
     - CONTACTS _____ 76 VOLTS DC CAPABLE OF WITHSTANDING 90 VOLTS DC AT NO LOAD
   - (B) MAXIMUM COIL VOLTAGE __ 31 VOLTS DC FOR 30 MINUTES
   - (C) MAXIMUM COIL CURRENT __ 1.6 AMPS AT 31 VOLTS DC AND 77° F
   - (D) RATED CONTACT LOAD ____ 400 AMPS; CAPABLE OF OPERATING (OPENING) AT MOMENTARY LOAD OF 600 AMPS
   - (E) PICKUP VOLTAGE (MAXIMUM) 18 VOLTS DC
   - (F) DROPOUT VOLTAGE _____ 9 VOLTS DC +0, −7 VOLTS
   - (G) RATED DUTY CYCLE _____ 30 MIN ON, 3 HRS OFF
   - (H) MINIMUM OPERATING CYCLES 250
   - (I) MAXIMUM WEIGHT _____ 2.5 LB

3. UNLESS OTHERWISE SPECIFIED HEREON, THE REQUIREMENTS OF MIL-R-00045 APPLY.

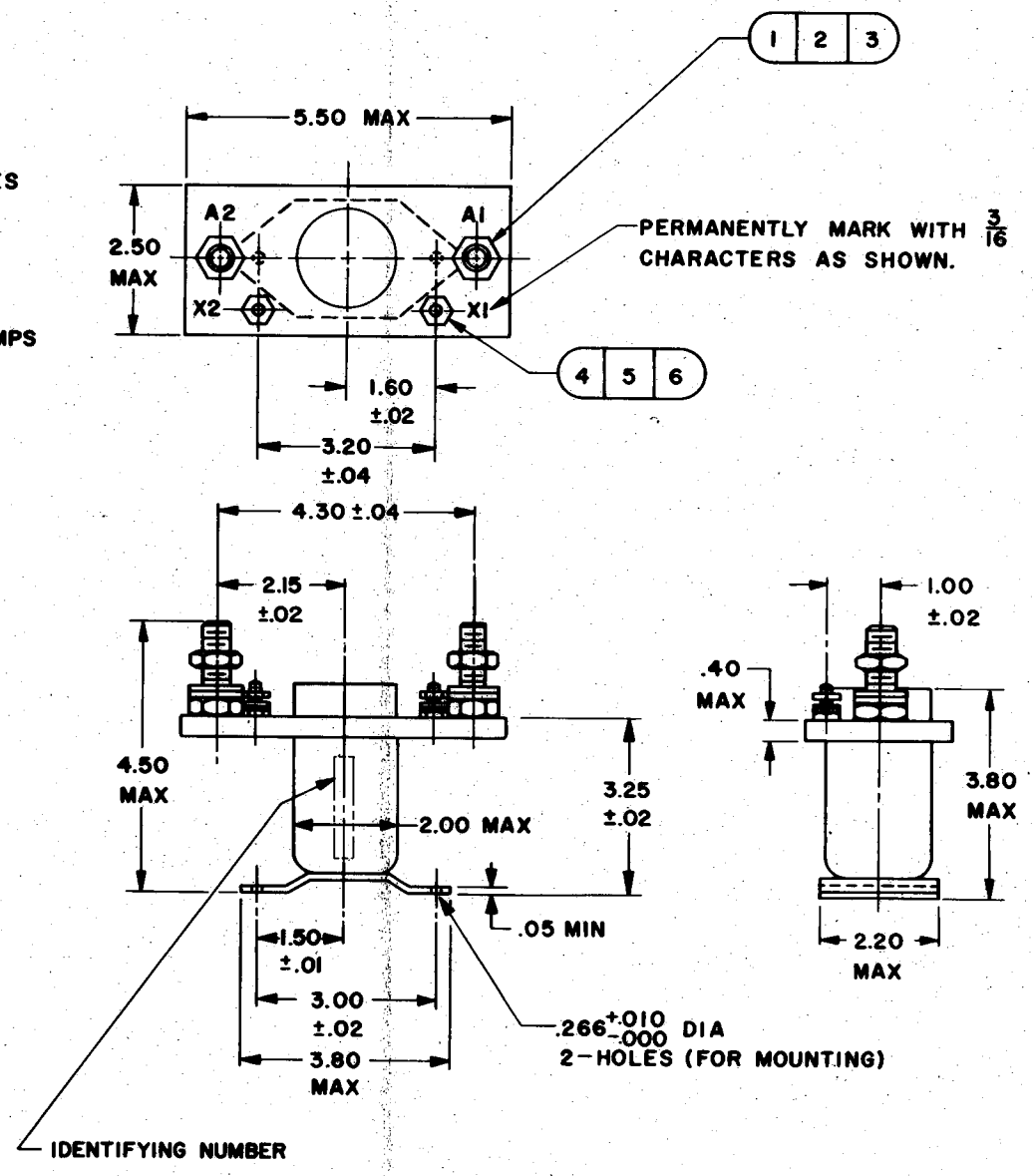
4. SUGGESTED SOURCES OF SUPPLY:

   K.C. JONES
   1022 PRATT STREET
   WEST MEADOW, OHIO
   VENDOR'S ITEM NO. 292-4888G1

   EUCLID ELECTRIC LTD.
   6242 GAY STREET
   MAUMEE, KANSAS
   VENDOR'S ITEM NO. 706HT4



SCHEMATIC DIAGRAM



REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPROVED

FIND NO.	DESCRIPTION OF VENDORS PARTS	NO. REQD	GOVT ALTERNATIVE
6	WASHER, 5/32 ID BRASS	4	MS90007-22
5	NUT, BRASS .138 - 32 UNC	4	MS00009-57
4	STUD, BRASS .138 - 32 UNC	2	MS00001-62
3	WASHER, 9/16 ID BRASS	4	MS0007-12
2	NUT, BRASS 1/2 -20 UNF	4	MS00009-11
1	STUD, BRASS 1/2 -20 UNF	2	MS00001-16

REPLACEMENTS FOR EXTERNAL (LOOSE) FITTINGS

RELAY, SOLENOID
SPECIFICATION CONTROL DRAWING

SIZE	CODE IDENT NO.	DRAWING NO.
	47XX1	6639XX2

SCALE | SHEET

FIGURE 7. Sample Specification Control Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard

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NOTES

1. FOR DEFINITION OF TOP CENTER LINE LAUNCH SEE NOTE 1 ON DWG NO. 0000224
2. THESE DIMENSIONS DEFINE THE MAXIMUM ALLOWABLE HANDLING AREA. MISSILE HARD SPOT MAXIMUM ALLOWABLE PRESSURE (DUE TO HANDLING ATTACHMENT CLAMPING PRESSURE AND SHOCK LOADS DURING HANDLING OR STORAGE) SHALL NOT EXCEED
   - (a) 500 PSI
   - (b) 500 PSI
   - (c) 600 PSI
   - (d) 415 PSI
   - (e) 415 PSI
   - (f) 2200 PSI
3. BOOSTER SUPPORT AREA (IN ADDITION TO LAUNCHING AND HANDLING SHOES) MAXIMUM ALLOWABLE PRESSURE (DUE TO HANDLING ATTACHMENT CLAMPING PRESSURE AND SHOCK LOADS) SHALL NOT EXCEED 400 PSI
4. FOR ELECTRICAL INTERCONNECTION COORDINATION DRAWING, SEE DWG NO. 0000225
5. FOR HYDRAULIC-PNEUMATIC COORDINATION DRAWING, SEE DWG NO. 0000226
6. FOR SYSTEM INSTALLATION COORDINATION DRAWING, SEE DWG NO. 0000227

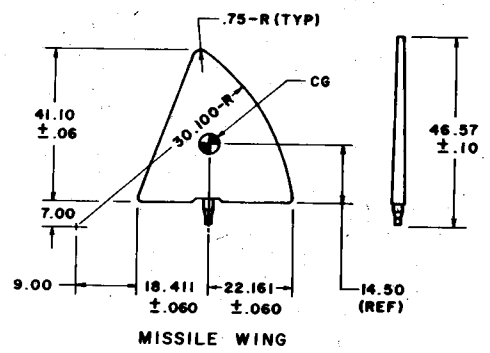
MAXIMUM VERTICAL DEFLECTION (INCHES) SUPPORTED BY BOOSTER SHOES

CONDITION	STATION						
	0.000	15.780	81.300	121.000	197.630	234.000	255.000
MECHANICAL ASSEMBLY TOLERANCES		1.080	0.846	0.604	0.419	0.204	

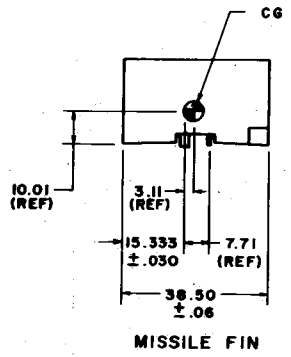
ANGULAR MISALIGNMENT OF AERODYNAMIC SURFACES

ITEM	MISSILE WING	MISSILE FIN	BOOSTER FIN
TOP CENTER LINE OF LAUNCH TO BOOSTER TORQUE KEY	0°9'	0°9'	0°9'
TORQUE KEY TO MISSILE KEYWAY	0°5'	0°5'	——
TORQUE KEY TO SURFACES	——	——	0°54'
KEYWAY TO SURFACES	1°40'	1°40'	——
TOTAL	1°54'	1°54'	1°3'

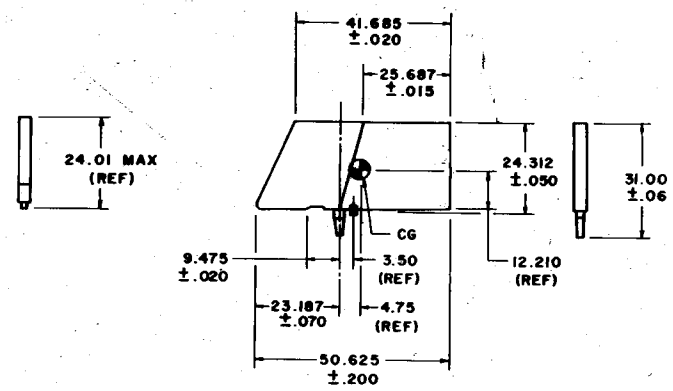
REVISIONS				
ZONE	LTR	DESCRIPTION	DATE	APPROVED



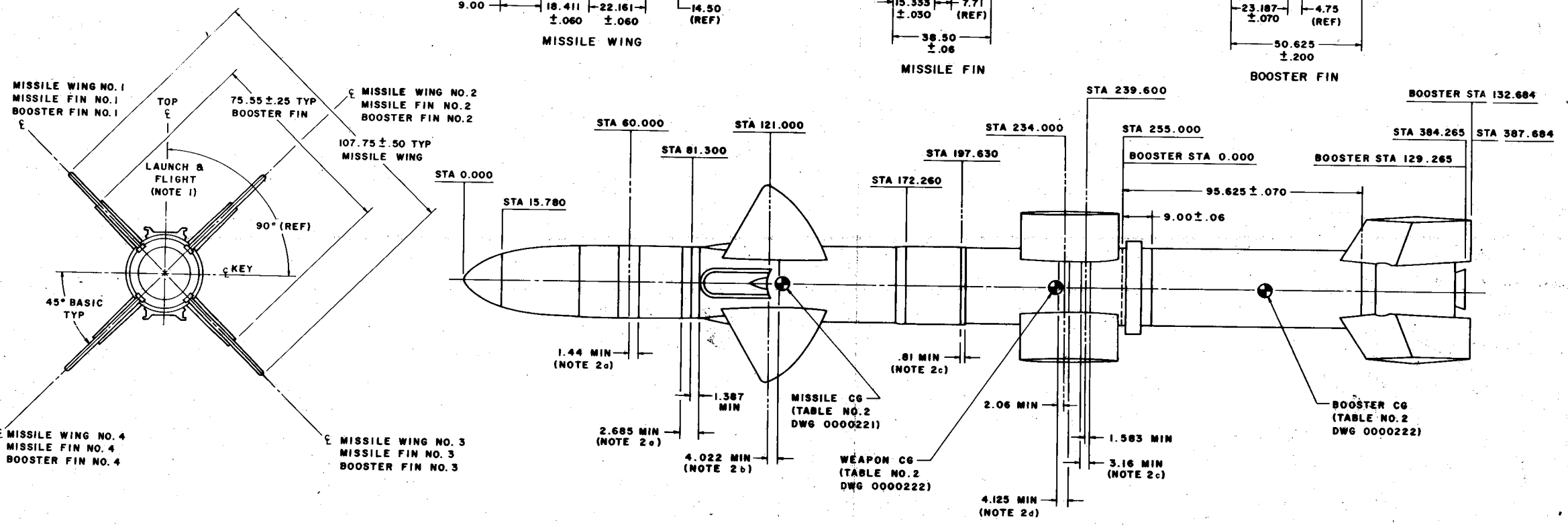
MISSILE WING



MISSILE FIN



BOOSTER FIN



COORDINATION DRAWING

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ON FRACTIONS DECIMALS ANGLES ±.02	MISSILE CONFIGURATION OUTLINE
MATERIAL	SIZE / CODE IDENT NO. / DRAWING NO.
	SCALE / SHEET

FIGURE 8. Sample Coordination Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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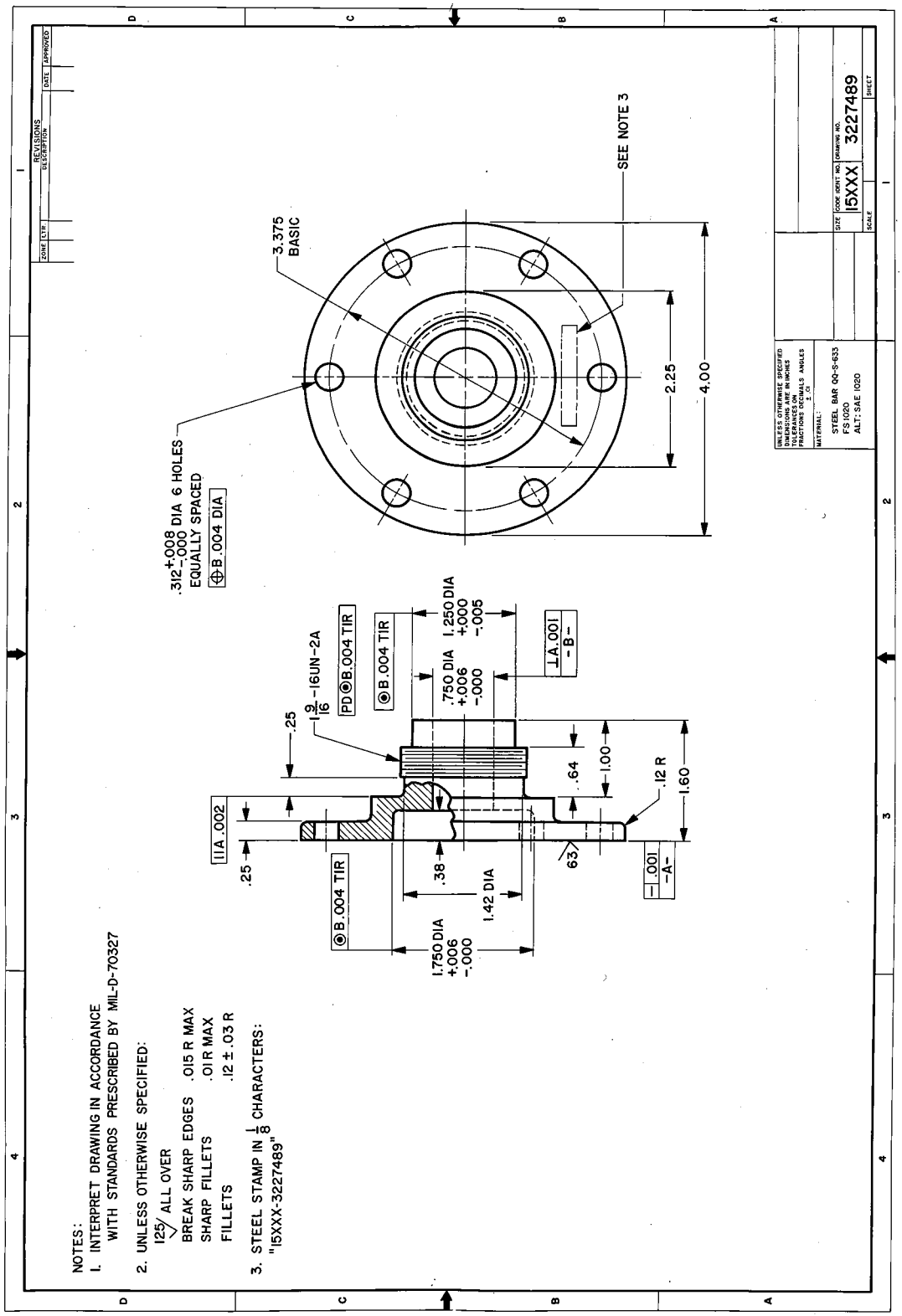


FIGURE 9. Sample Detail Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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Check the source to verify that this is the current version before use.

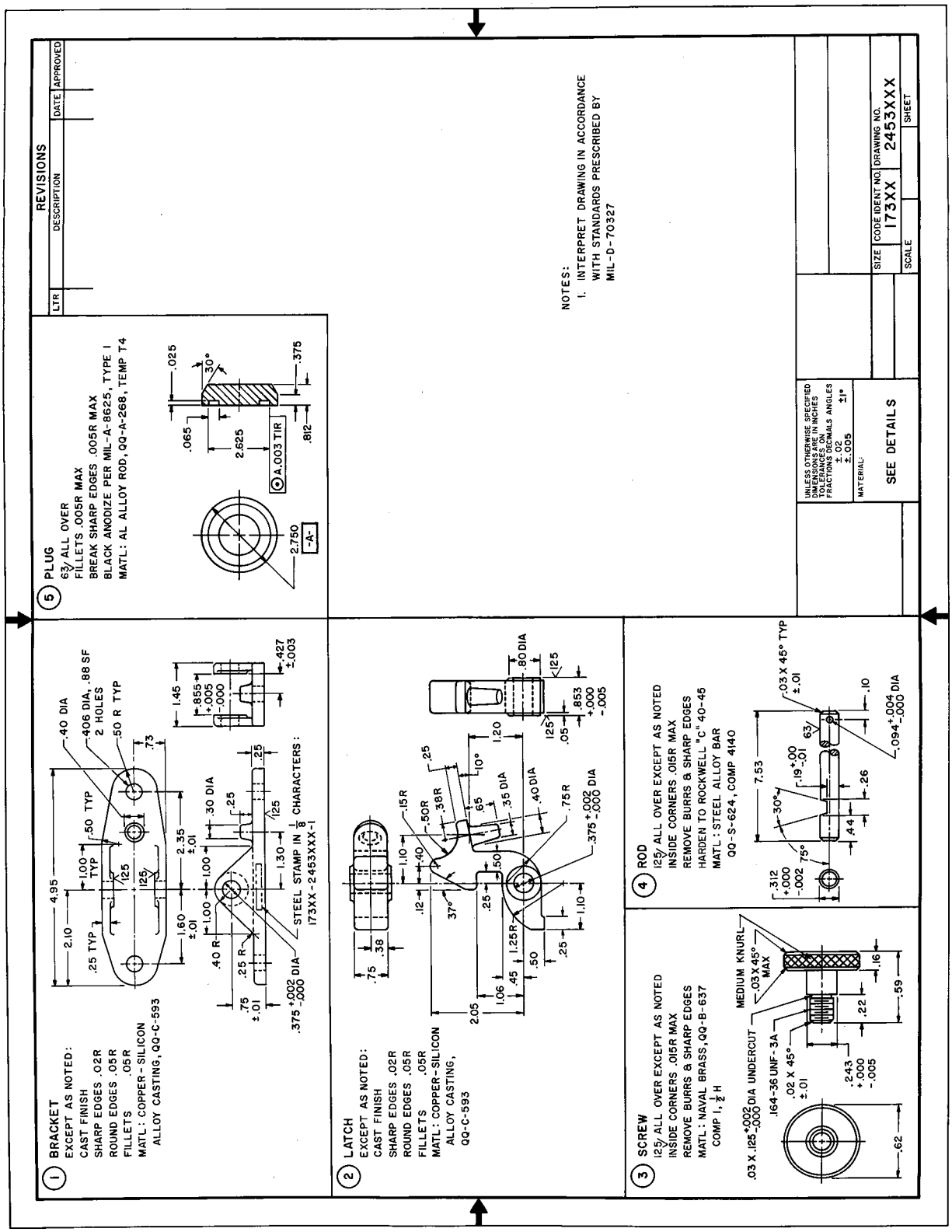


FIGURE 10. Sample Multi Detail Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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Check the source to verify that this is the current version before use.

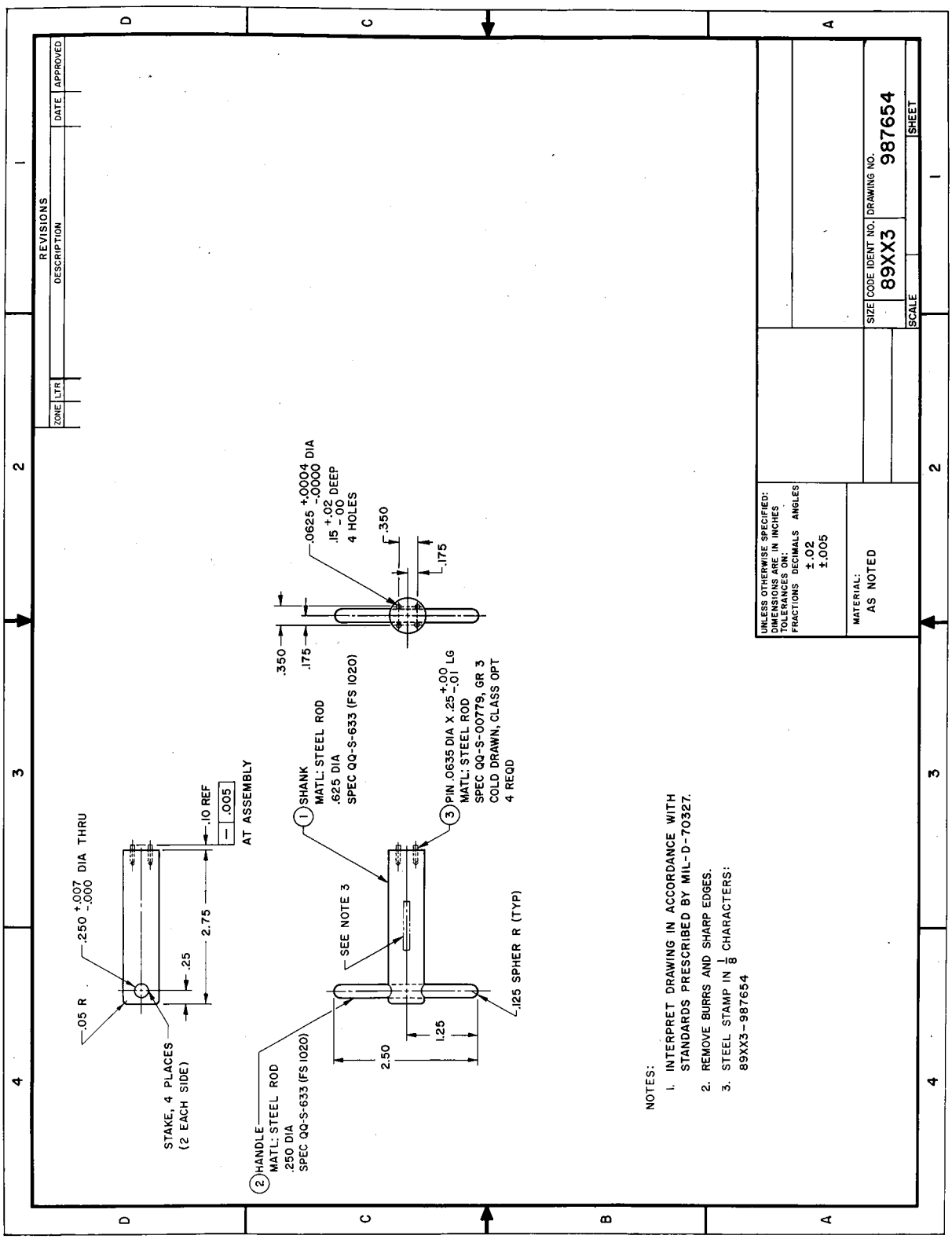


FIGURE 11. Sample Detail Assembly Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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Check the source to verify that this is the current version before use.

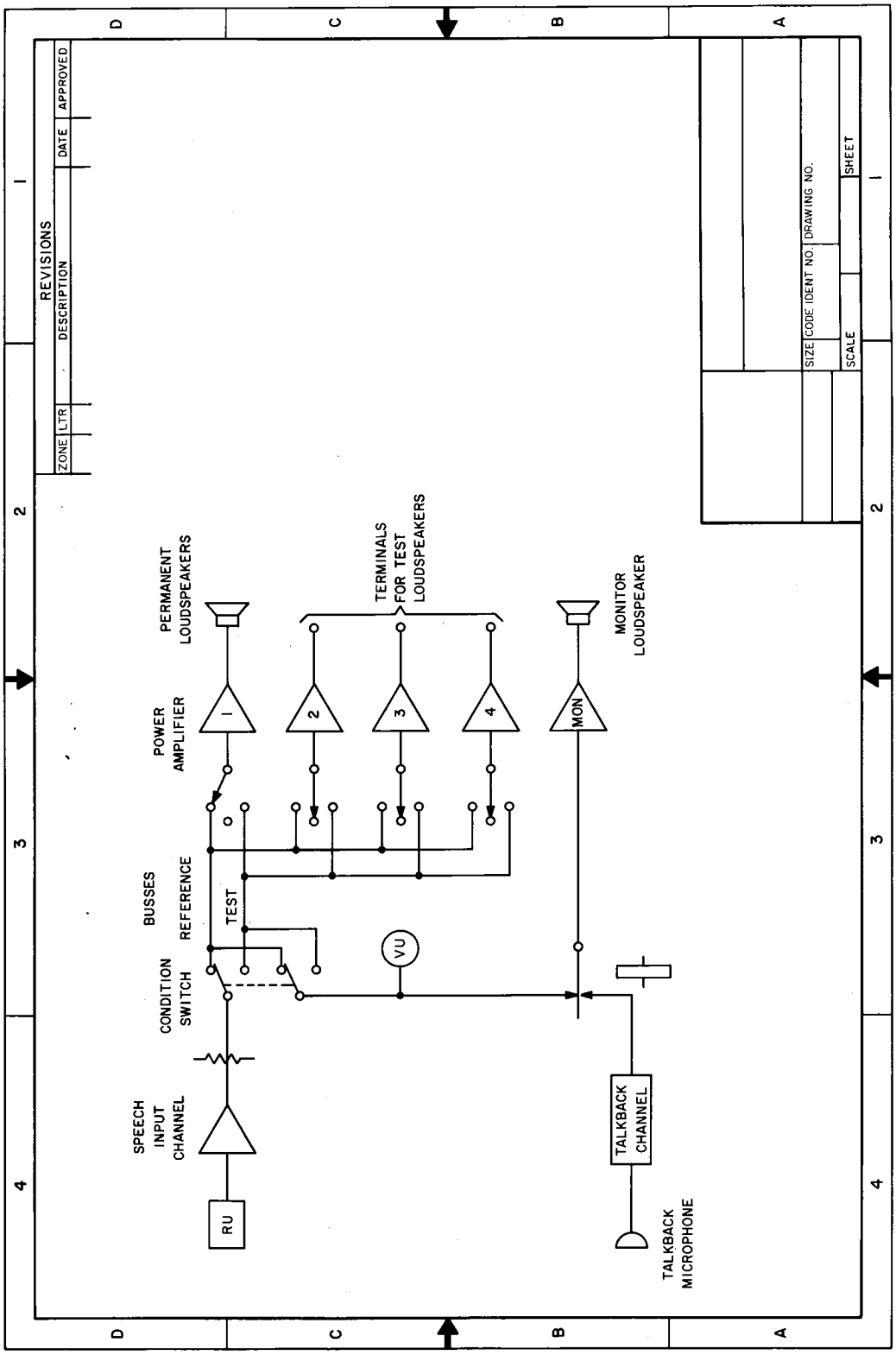
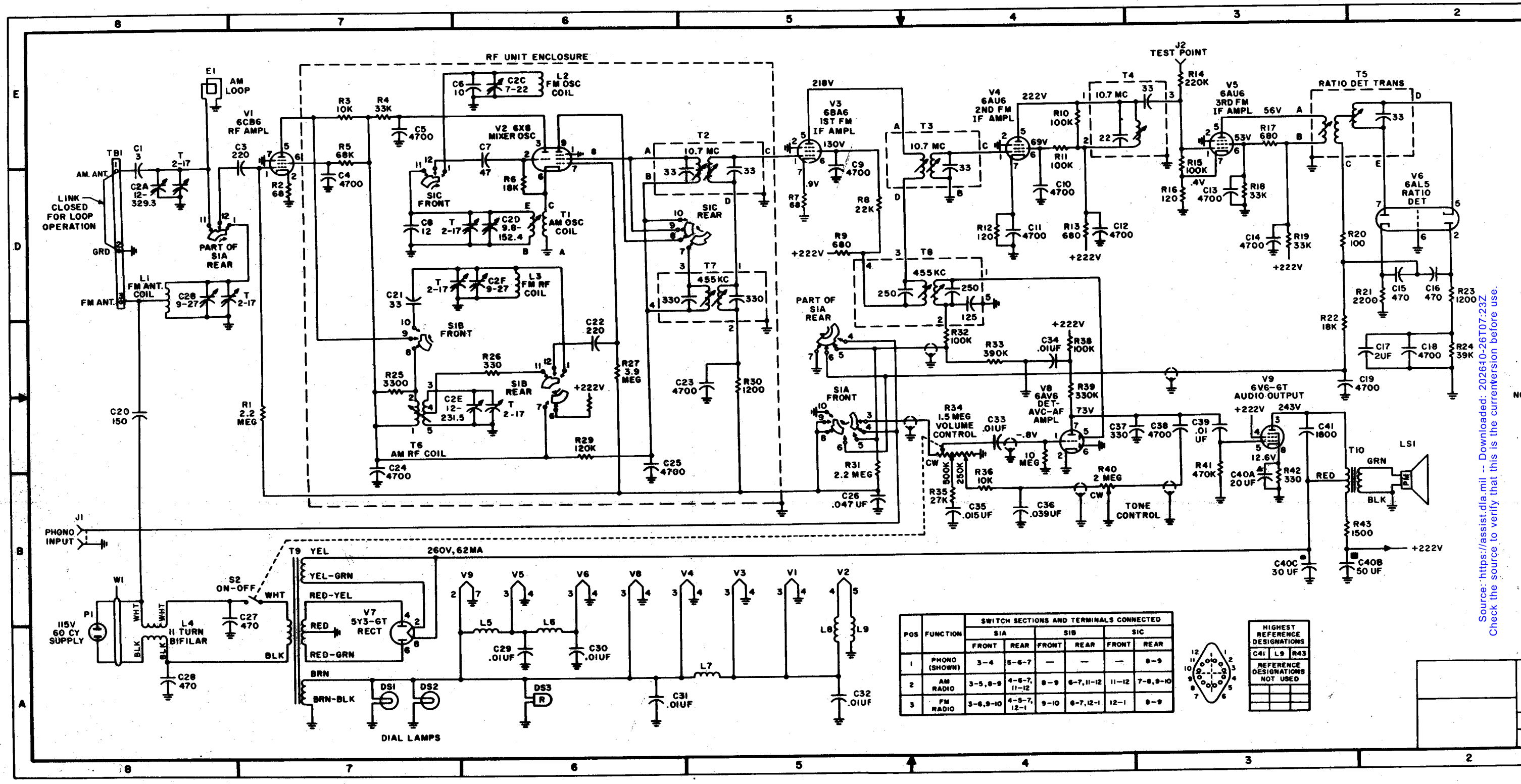


FIGURE 12. Sample Single Line Diagram Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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RF UNIT ENCLOSURE
E1 AM LOOP
V1 6CB6 RF AMPL
V2 6X8 MIXER OSC
V3 6BA6 1ST FM IF AMPL
V4 6AU6 2ND FM IF AMPL
V5 6AU6 3RD FM IF AMPL
V6 6AL5 RATIO DET
V7 5Y3-GT RECT
V8 6AV6 DET-AVC-AF AMPL
V9 6V6-GT AUDIO OUTPUT
T1 AM OSC COIL
T2 10.7 MC
T3 10.7 MC
T4 10.7 MC
T5 RATIO DET TRANS
T6 AM RF COIL
T7 455KC
T8 455KC
T9 260V, 62MA
T10
L1 FM ANT. COIL
L2 FM OSC COIL
L3 FM RF COIL
L4 11 TURN BIFILAR
L5
L6
L7
L8
L9
J1 PHONO INPUT
J2 TEST POINT
P1 115V 60 CY SUPPLY
W1
S1A
S1B
S1C
S2 ON-OFF
DS1
DS2
DS3
DIAL LAMPS
LINK CLOSED FOR LOOP OPERATION
PART OF S1A REAR
SIA FRONT
SIA REAR
SIB FRONT
SIB REAR
SIC FRONT
SIC REAR
R1 2.2 MEG
R2 68
R3 10K
R4 33K
R5 68K
R6 18K
R7 68
R8 22K
R9 680
R10 100K
R11 100K
R12 120
R13 680
R14 220K
R15 100K
R16 120
R17 680
R18 33K
R19 33K
R20 100
R21 2200
R22 18K
R23 1200
R24 39K
R25 3300
R26 330
R27 3.9 MEG
R29 120K
R30 1200
R31 2.2 MEG
R32 100K
R33 390K
R34 1.5 MEG VOLUME CONTROL
R35 27K
R36 10K
R38 100K
R39 330K
R40 2 MEG
R41 470K
R42 330
R43 1500
C1 3
C2A 12-329.3
C2C 7-22
C2D 9.8-152.4
C2E 12-231.5
C2F 9-27
C3 220
C4 4700
C5 4700
C6 10
C7 47
C8 12
C9 4700
C10 4700
C11 4700
C12 4700
C13 4700
C14 4700
C15 470
C16 470
C17 2UF
C18 4700
C19 4700
C20 150
C21 33
C22 220
C23 4700
C24 4700
C25 4700
C26 .047 UF
C27 470
C28 470
C28 9-27
C29 .01UF
C30 .01UF
C31 .01UF
C32 .01UF
C33 .01UF
C34 .01UF
C35 .015UF
C36 .039UF
C37 330
C38 4700
C39 .01 UF
C40A 20 UF
C40B 50 UF
C40C 30 UF
C41 1800
T 2-17
T 9-27
T 2-17
T 2-17
T 2-17

NOTES

1. DATA FOR WIRE NUMBERS 1 THROUGH 81 ARE STATED IN TABLE ON DWG 0000809.
2. USE QQ-S-571, SN60 FOR ALL SOLDERED CONNECTIONS.
3. PARTIAL REFERENCE DESIGNATIONS ARE SHOWN. PREFIX THE PART DESIGNATION WITH SUB-ASSEMBLY DESIGNATION.

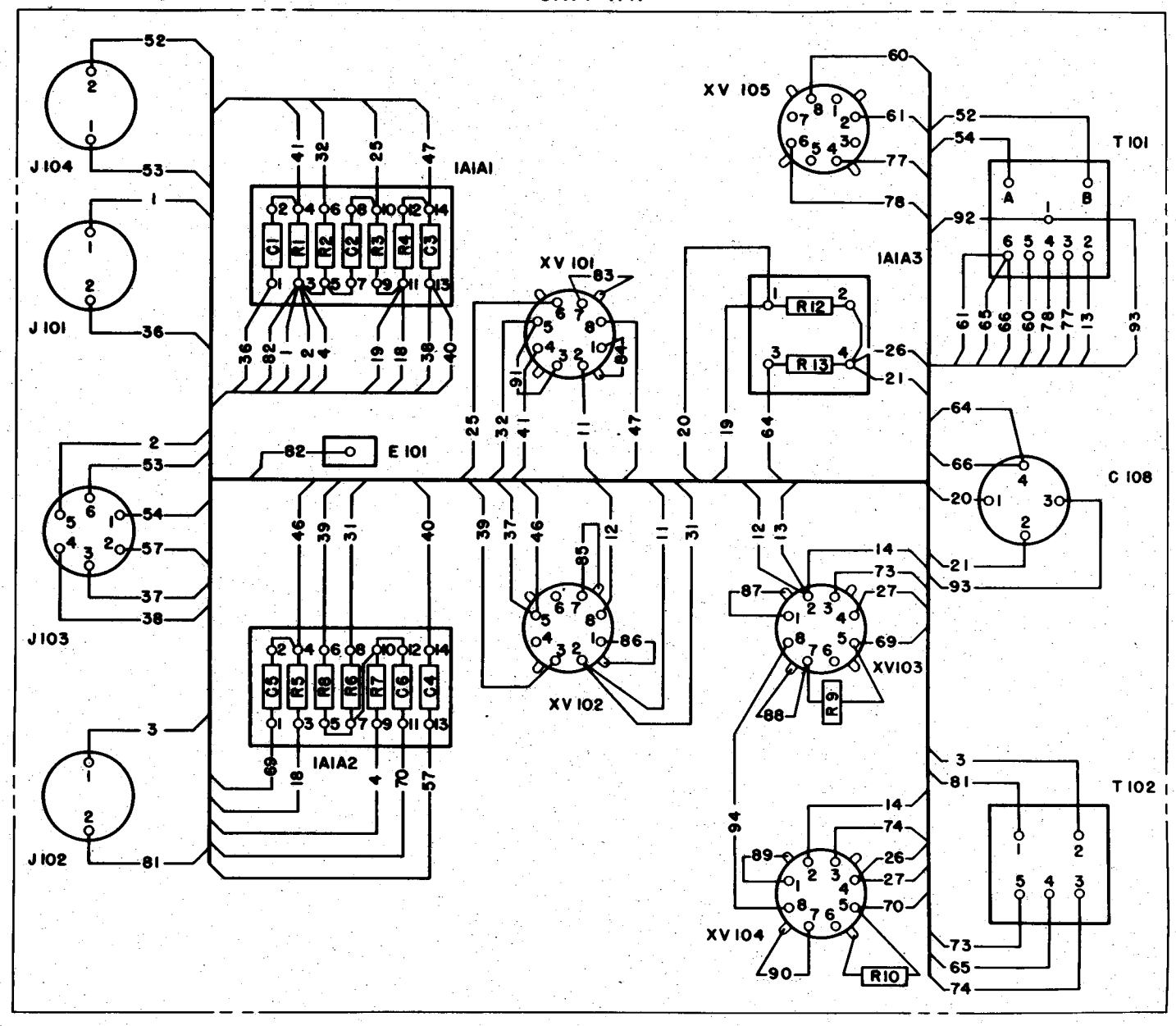
REFERENCES

1. LIST OF MATERIAL, LM 0000810.
2. SCHEMATIC DIAGRAM, DWG 0000812.
3. WIRING HARNESS, BRANCHED, DWG 0000809.
4. ASSEMBLY, DWG 0000813.

RUNNING LIST

WIRE NO.	BODY	FROM	TO
82	BARE	TB101-3	E101
83	BARE	XV101-7	GRD LUG
84	BARE	XV101-1	GRD LUG
85	BARE	XV102-7	GRD LUG
86	BARE	XV102-1	GRD LUG
87	BARE	XV103-1	GRD LUG
88	BARE	XV103-7	GRD LUG
89	BARE	XV104-1	GRD LUG
90	BARE	XV104-7	GRD LUG
91	INSULATED	XV101-3	XV101-5

RESISTOR NO.	FROM	TO
R9	XV103-5	XV103-7
R10	XV104-5	GRD LUG



REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPROVED

SIZE | CODE IDENT NO. | DRAWING NO. | SCALE | SHEET

FIGURE 14. Sample Wiring Diagram Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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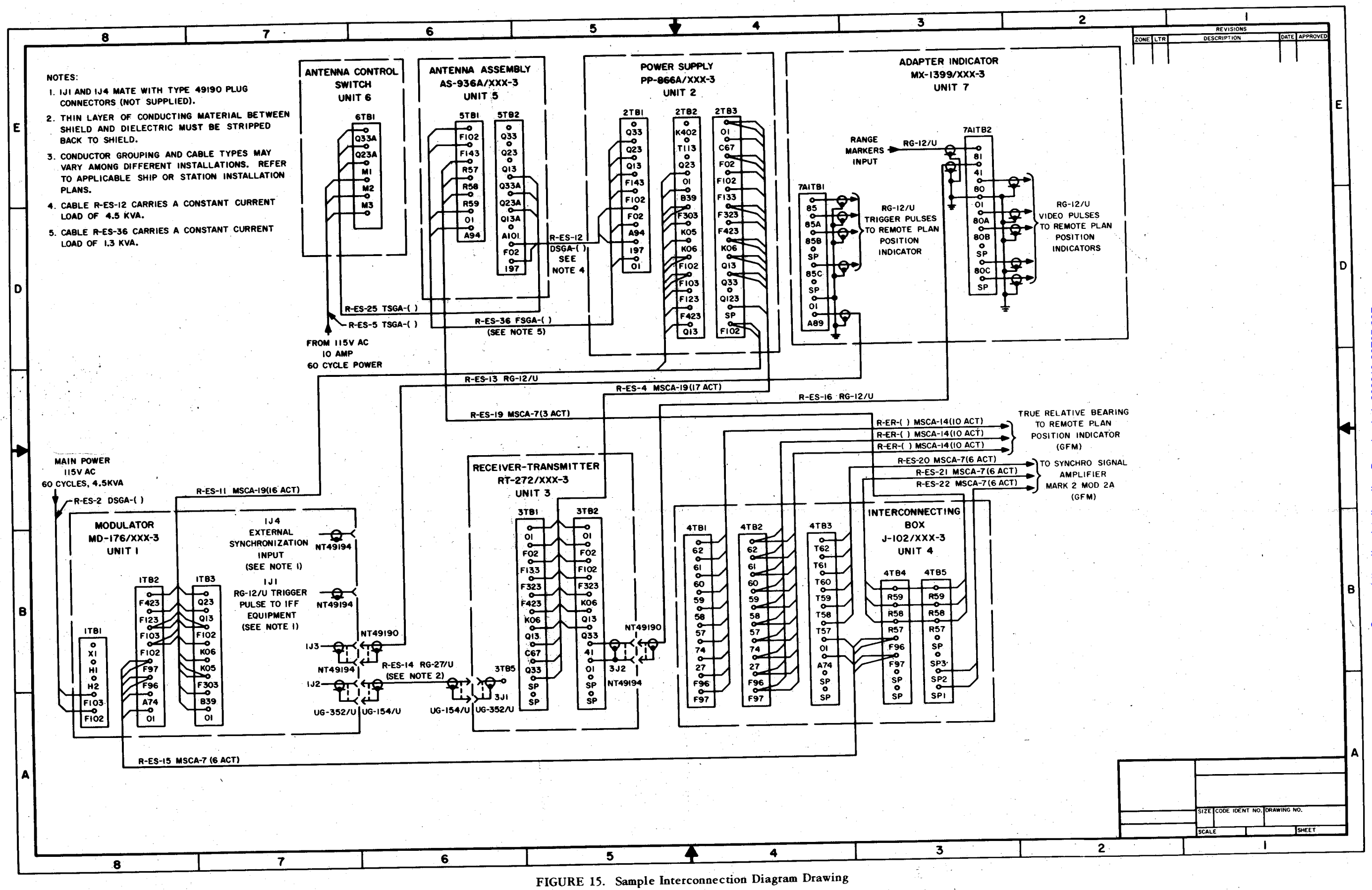


FIGURE 15. Sample Interconnection Diagram Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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Check the source to verify that this is the current version before use.

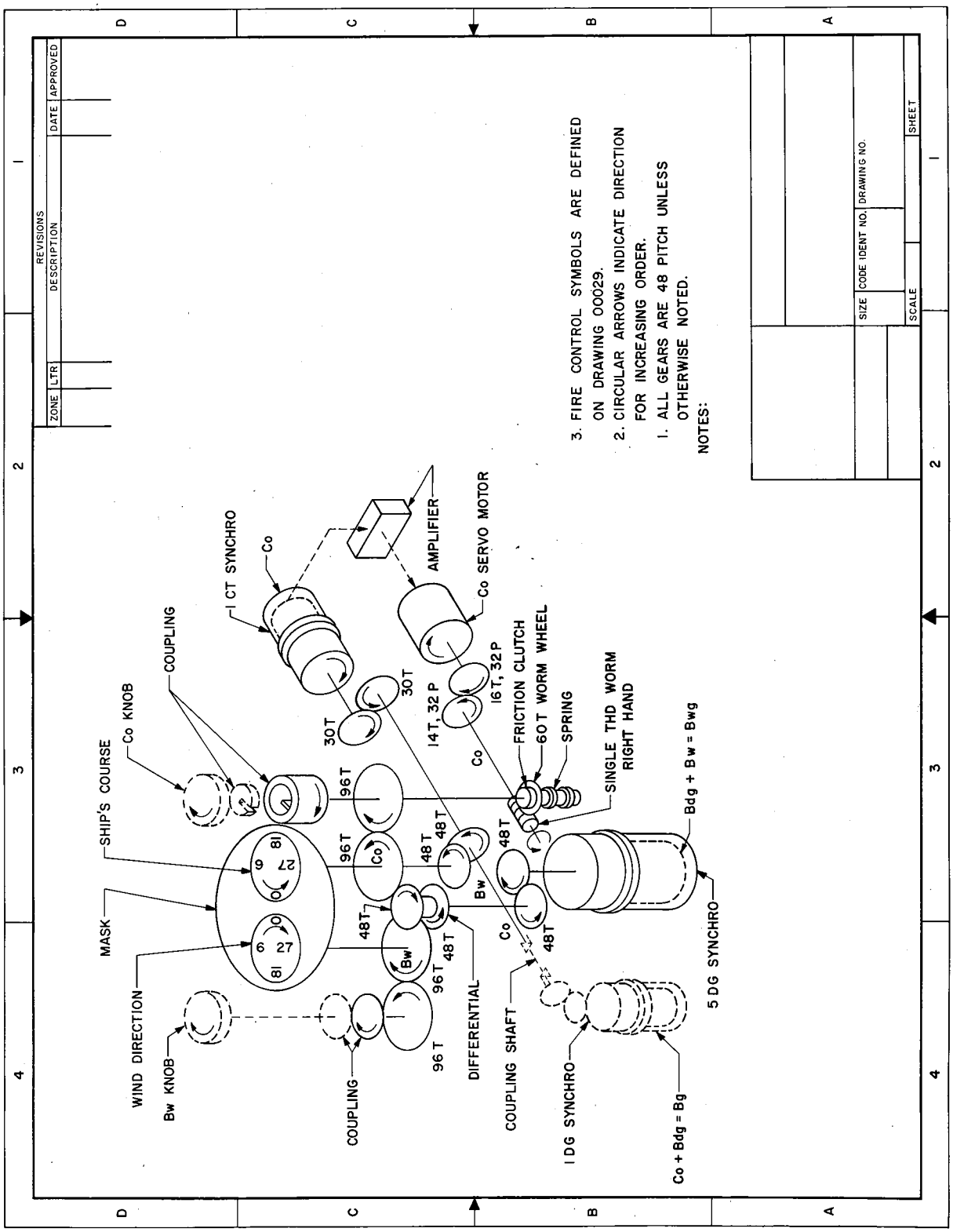


FIGURE 16. Sample Mechanical Schematic Diagram Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

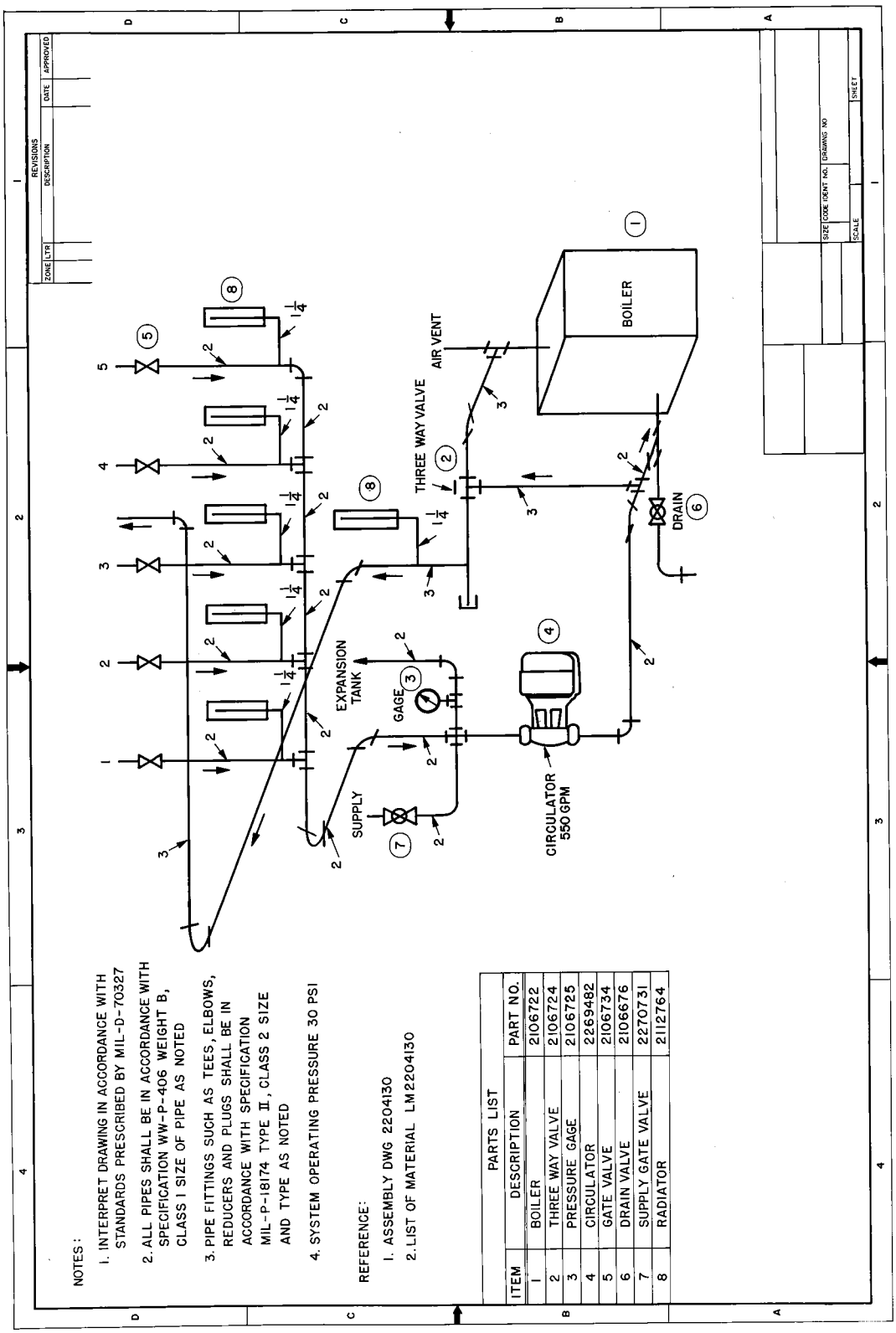


FIGURE 17. Sample Piping Diagram Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

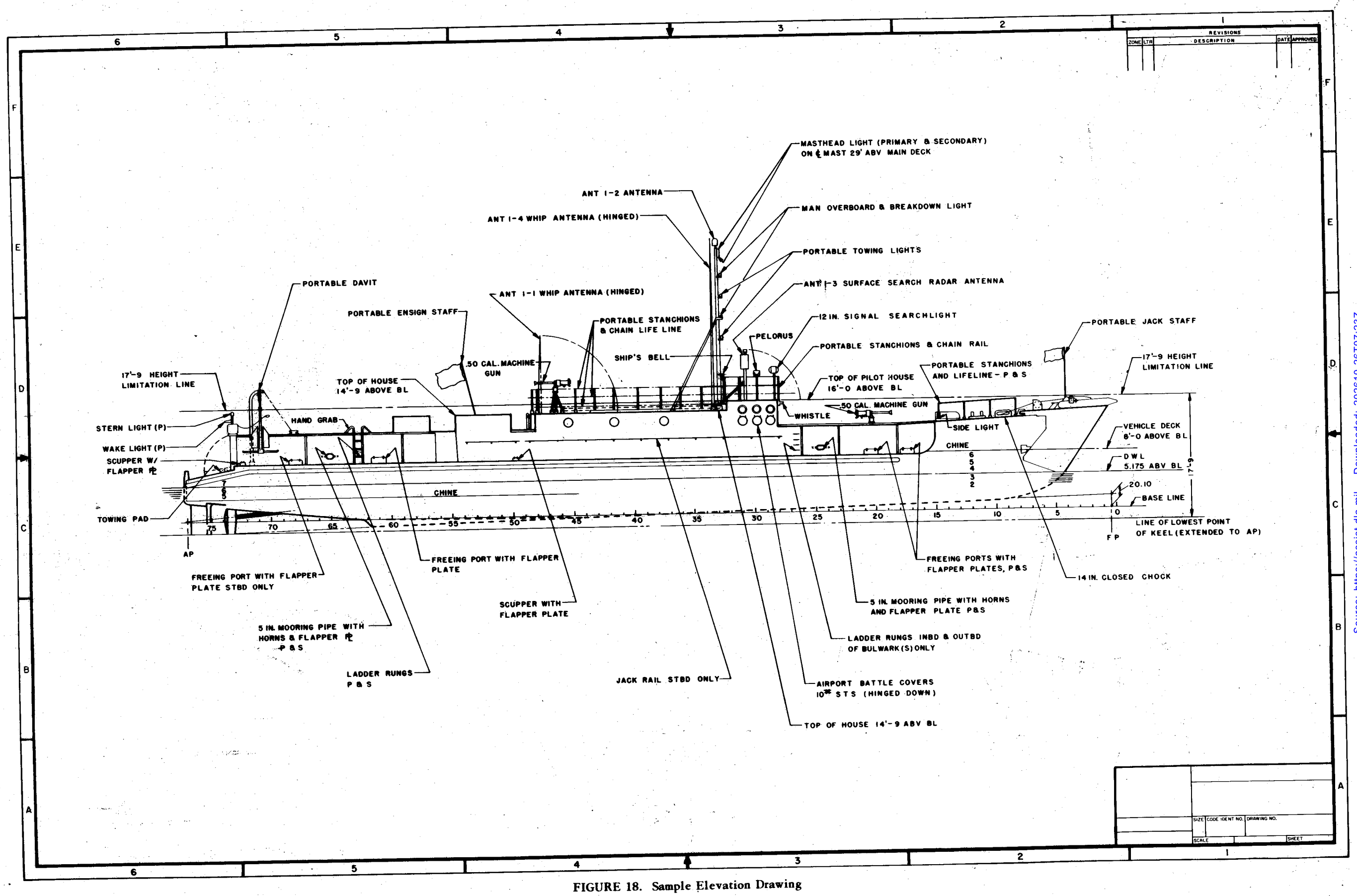


FIGURE 18. Sample Elevation Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

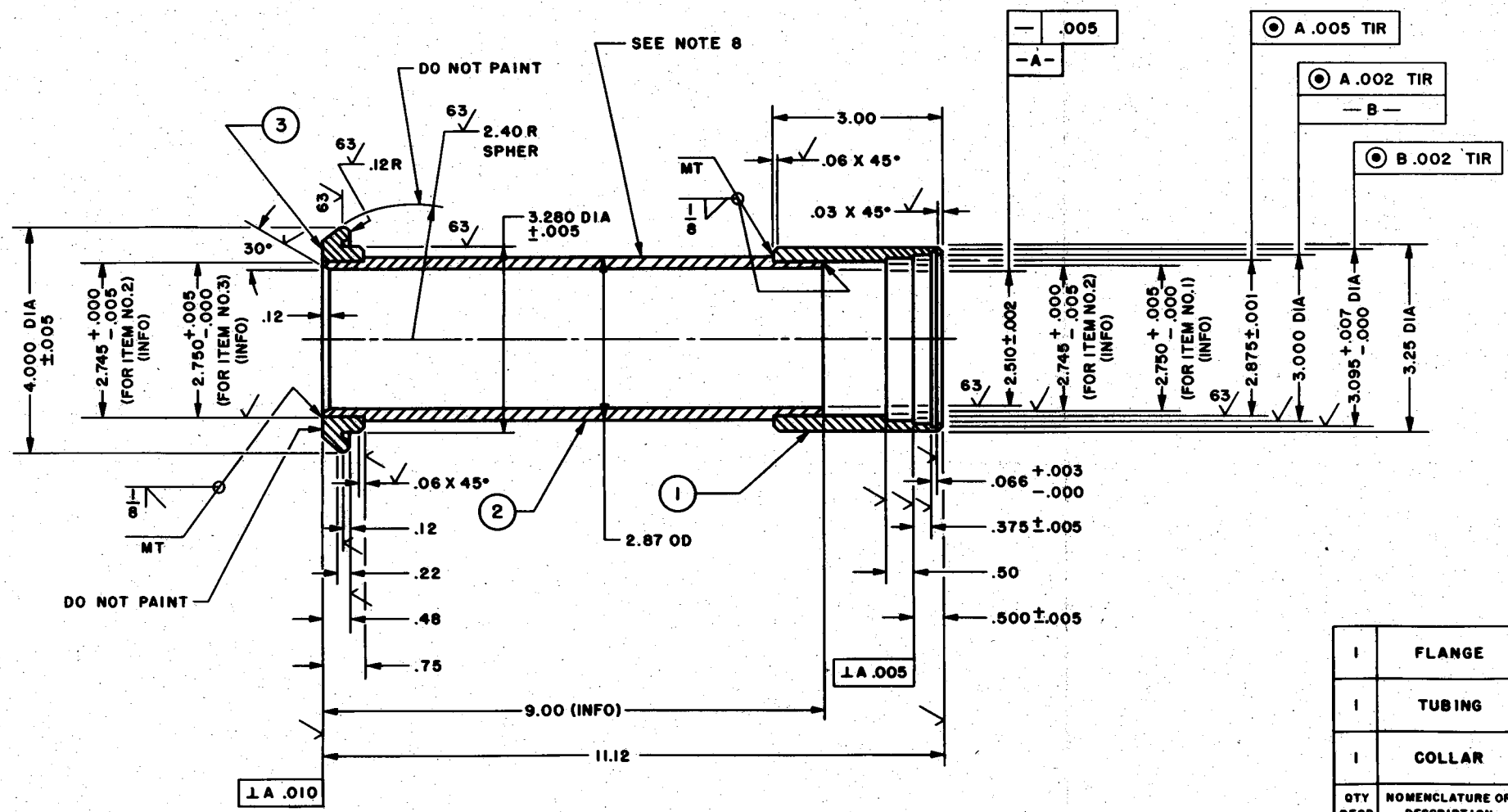
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

NOTES:

1. INTERPRET DRAWING IN ACCORDANCE WITH STANDARDS PRESCRIBED BY MIL-D-70327.
2. SURFACES MARKED $\sqrt{}$ TO HAVE $\sqrt{125}$ UNLESS OTHERWISE SPECIFIED: SURFACE ROUGHNESS AS ROLLED OR DRAWN INSIDE CORNERS .005 R MAX
3. WELDING SHALL BE ACCOMPLISHED BY THE MANUAL SHIELDED METAL ARC PROCESS WITH ELECTRODES CONFORMING TO MIL-E-00099, TYPE MIL-0010 OR MIL-0011.
4. FABRICATION PROCEDURES AND INSPECTION SHALL BE IN ACCORDANCE WITH MIL-W-21157, CLASS 2, EXCEPT THAT RADIOGRAPHIC INSPECTION IS WAIVED. STRESS RELIEVE AT 1175° F ±25°F FOR ONE HOUR.
5. THIS PART SHALL BE SUBJECTED TO A HYDROSTATIC PRESSURE OF 10 PSIA FOR A MINIMUM OF 30 MINUTES WITHOUT LEAKAGE.
6. UNLESS OTHERWISE SPECIFIED ALL DIA SHALL BE ⌀ A .020 TIR
7. PAINT ALL EXTERIOR SURFACES WITH TWO COATS MIL-P-1265, EXCEPT AS NOTED.
8. STEEL STAMP IN $\frac{1}{8}$ CHARACTERS: "75XX9-0000012"

REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPROVED



QTY REQD	NOMENCLATURE OR DESCRIPTION	CODE IDENT	IDENTIFYING NUMBER	SPECIFICATION	MATERIAL OR NOTE	ITEM NO.
1	FLANGE			QQ-S-633 C1020, CR	STEEL, CARBON	3
1	TUBING			MIL-T-16343, TYPE I	TUBING, STEEL	2
1	COLLAR			MIL-S-20166 TY. A, GR. M	STEEL, CARBON	1

LIST OF MATERIALS

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES
TOLERANCES ON: FRACTIONS ±.01 DECIMALS ANGLES

MATERIAL:

SIZE | CODE IDENT NO. 75XX9 | DRAWING NO. 0000012 | SCALE | SHEET

FIGURE 19. Sample Inseparable Assembly Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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Check the source to verify that this is the current version before use.

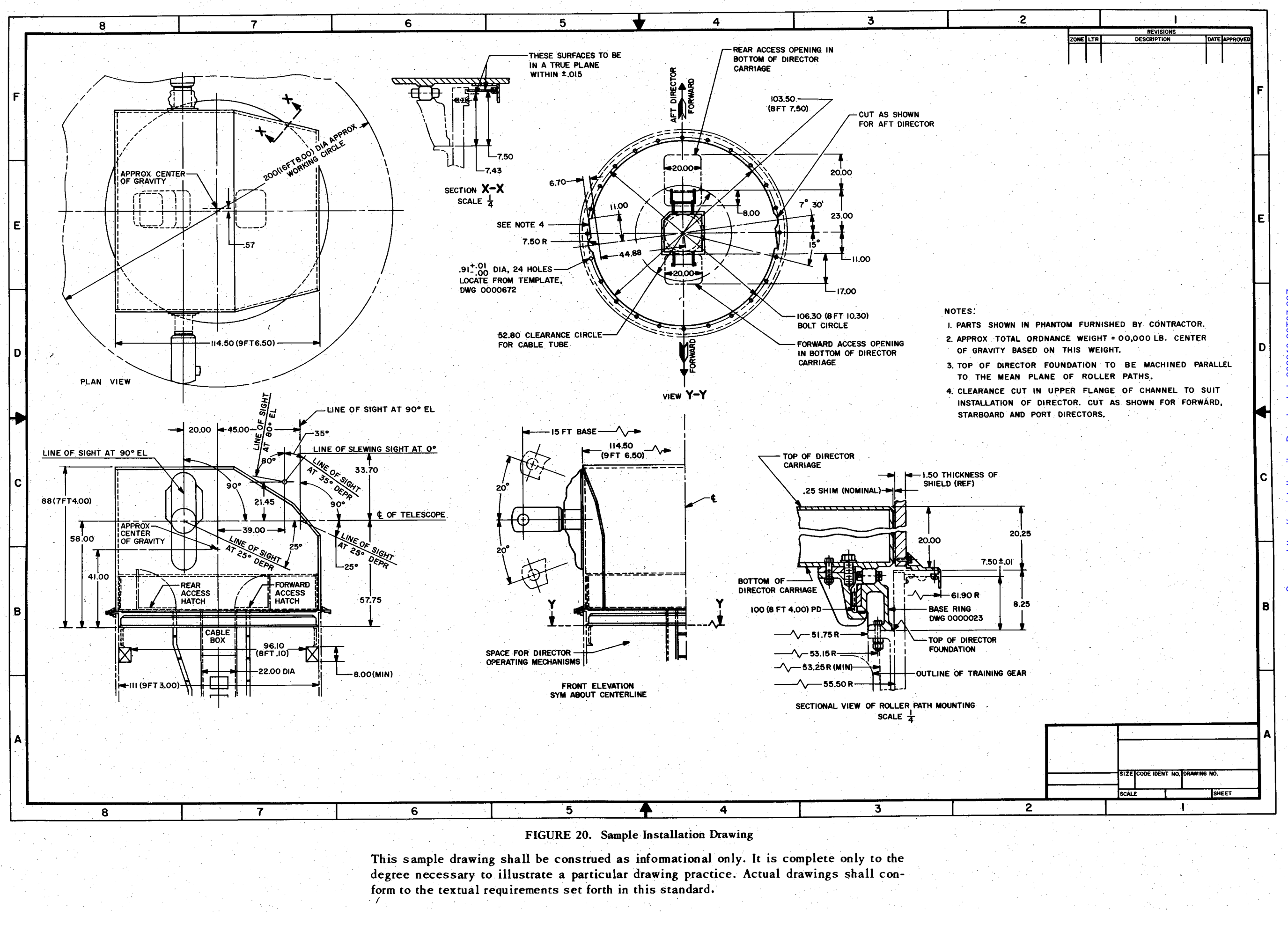
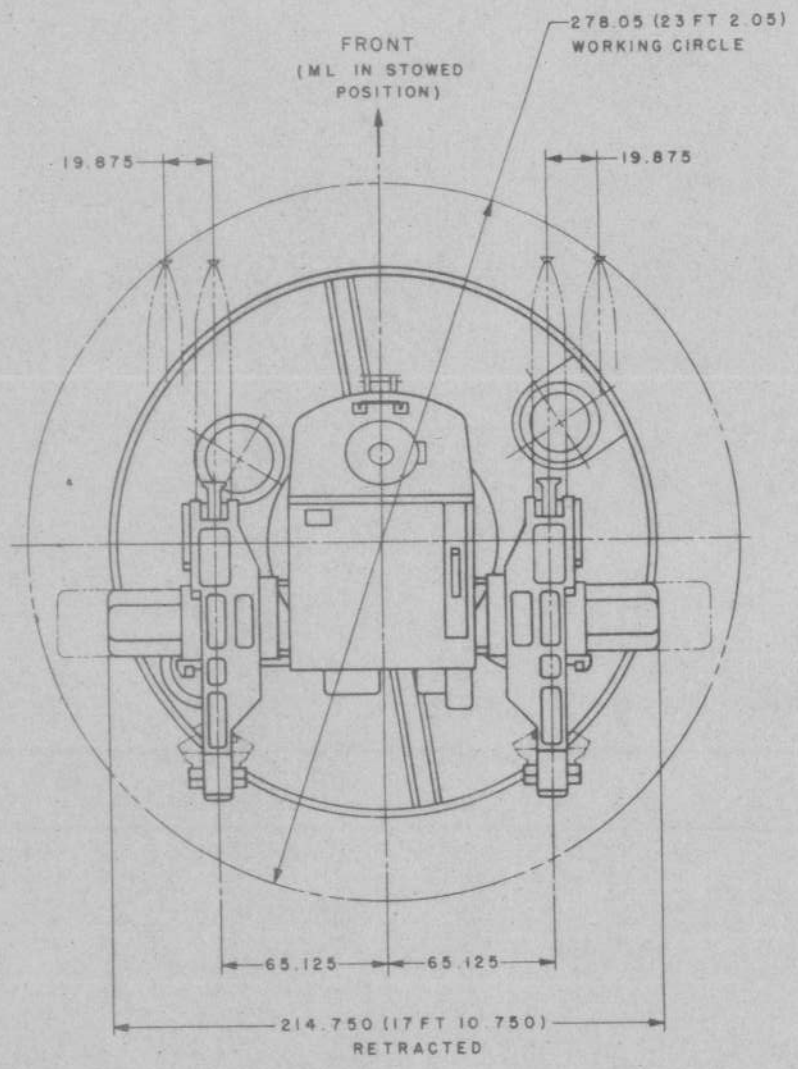


FIGURE 20. Sample Installation Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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REFERENCE DRAWINGS:
FOR GENERAL ARRANGEMENT, SEE DWG 0000009
FOR INSTALLATION DRAWING, SEE DWG 0000013



VIEW A-A
(ROTATED 90° CW FROM TRUE POSITION)

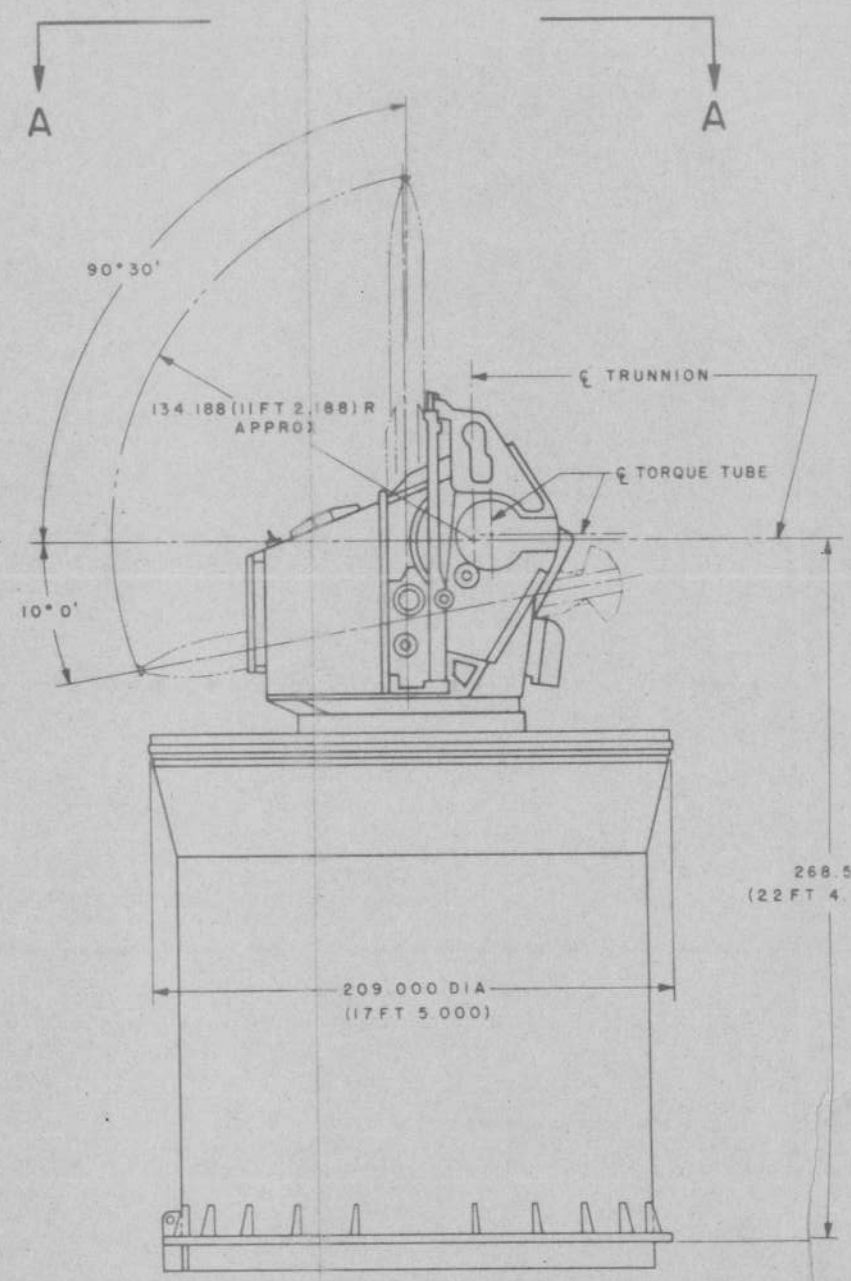
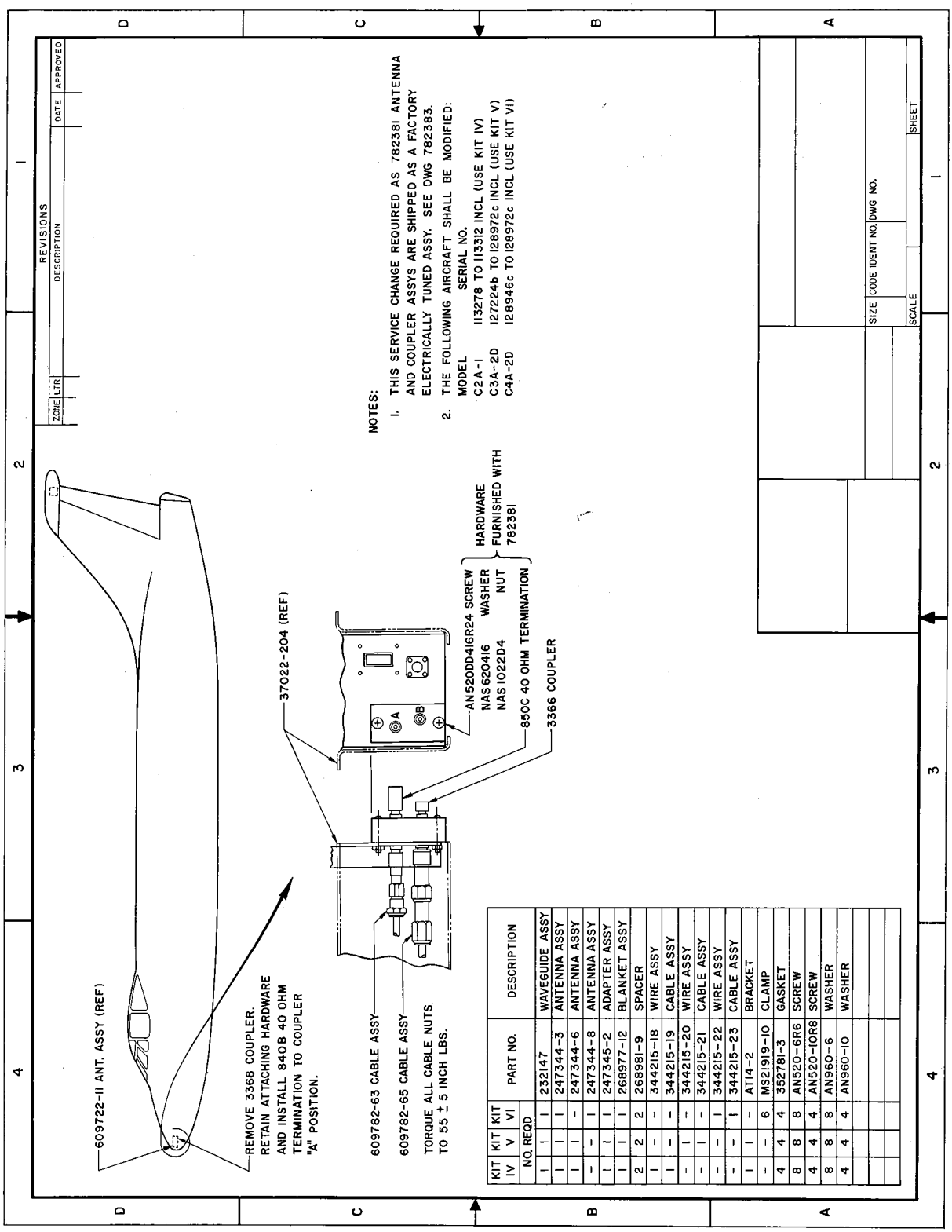


FIGURE 21. Sample Installation Control Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T07:23Z
Check the source to verify that this is the current version before use.



KIT IV	KIT V	KIT VI	NO. REQD	PART NO.	DESCRIPTION
1	1	1	1	232147	WAVEGUIDE ASSY
1	1	1	1	247344-3	ANTENNA ASSY
1	1	-	1	247344-6	ANTENNA ASSY
-	-	1	1	247344-8	ANTENNA ASSY
1	1	1	1	247345-2	ADAPTER ASSY
1	1	1	1	268977-12	BLANKET ASSY
2	2	2	2	268981-9	SPACER
1	-	-	1	344215-18	WIRE ASSY
1	-	-	1	344215-19	CABLE ASSY
-	1	-	1	344215-20	WIRE ASSY
-	1	-	1	344215-21	CABLE ASSY
-	-	1	1	344215-22	WIRE ASSY
-	-	1	1	344215-23	CABLE ASSY
1	1	-	1	AT14-2	BRACKET
-	-	6	6	MS21919-10	CLAMP
4	4	4	4	352781-3	GASKET
8	8	8	8	AN520-6R6	SCREW
4	4	4	4	AN520-10R8	SCREW
8	8	8	8	AN960-6	WASHER
4	4	4	4	AN960-10	WASHER

FIGURE 22. Sample Kit Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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Check the source to verify that this is the current version before use.

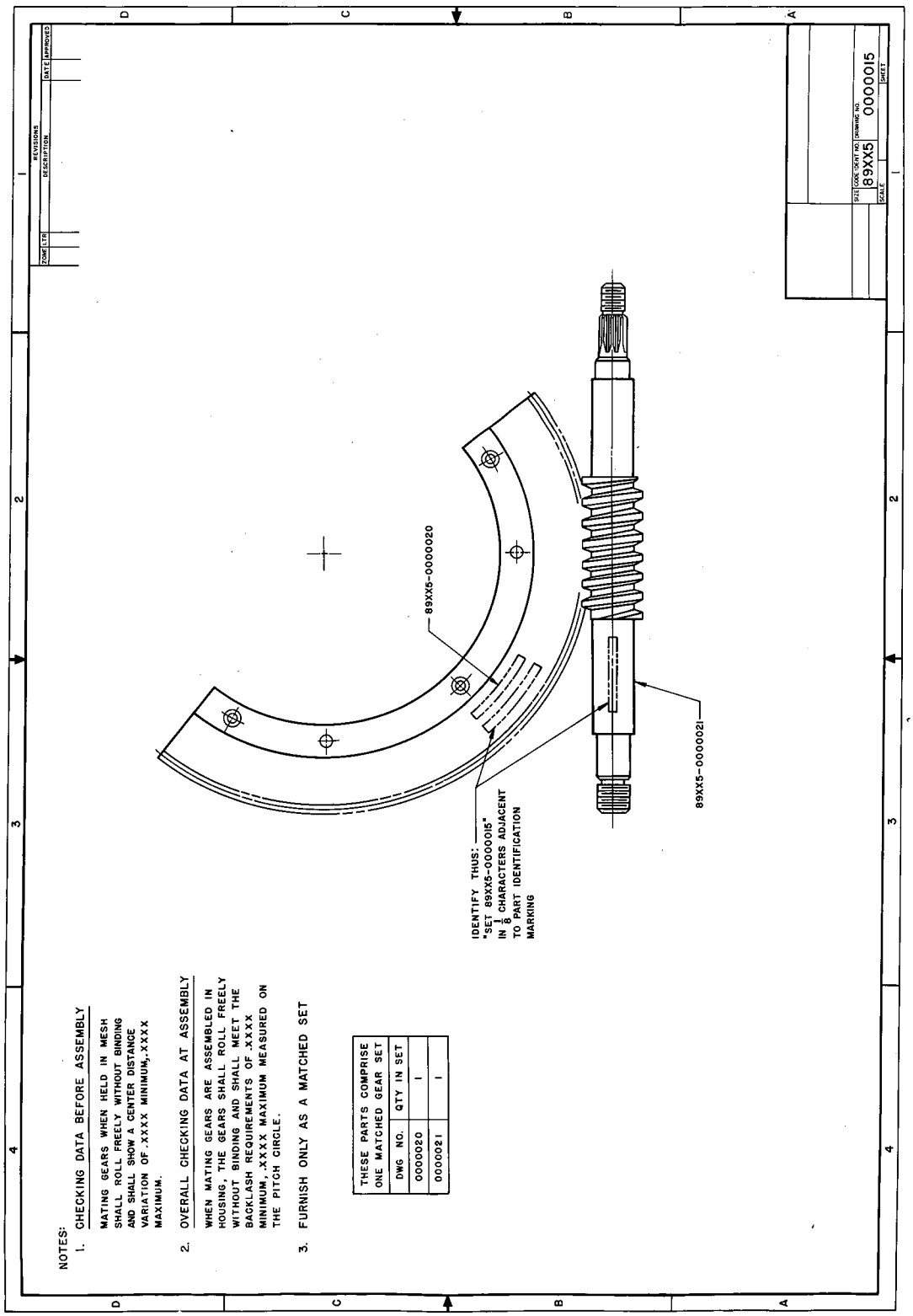
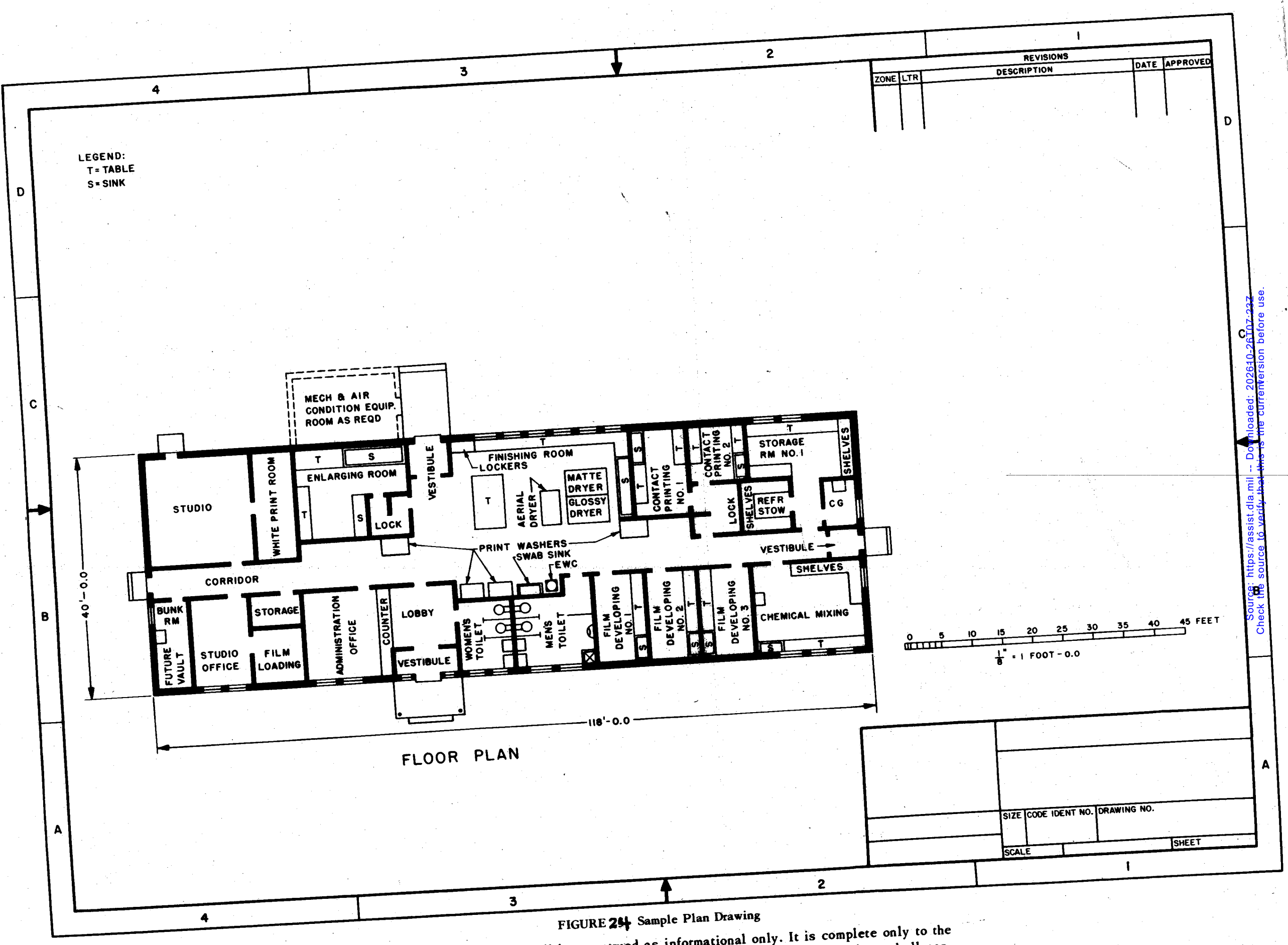


FIGURE 23. Sample Matched Parts Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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FLOOR PLAN

FIGURE 2~~4~~ Sample Plan Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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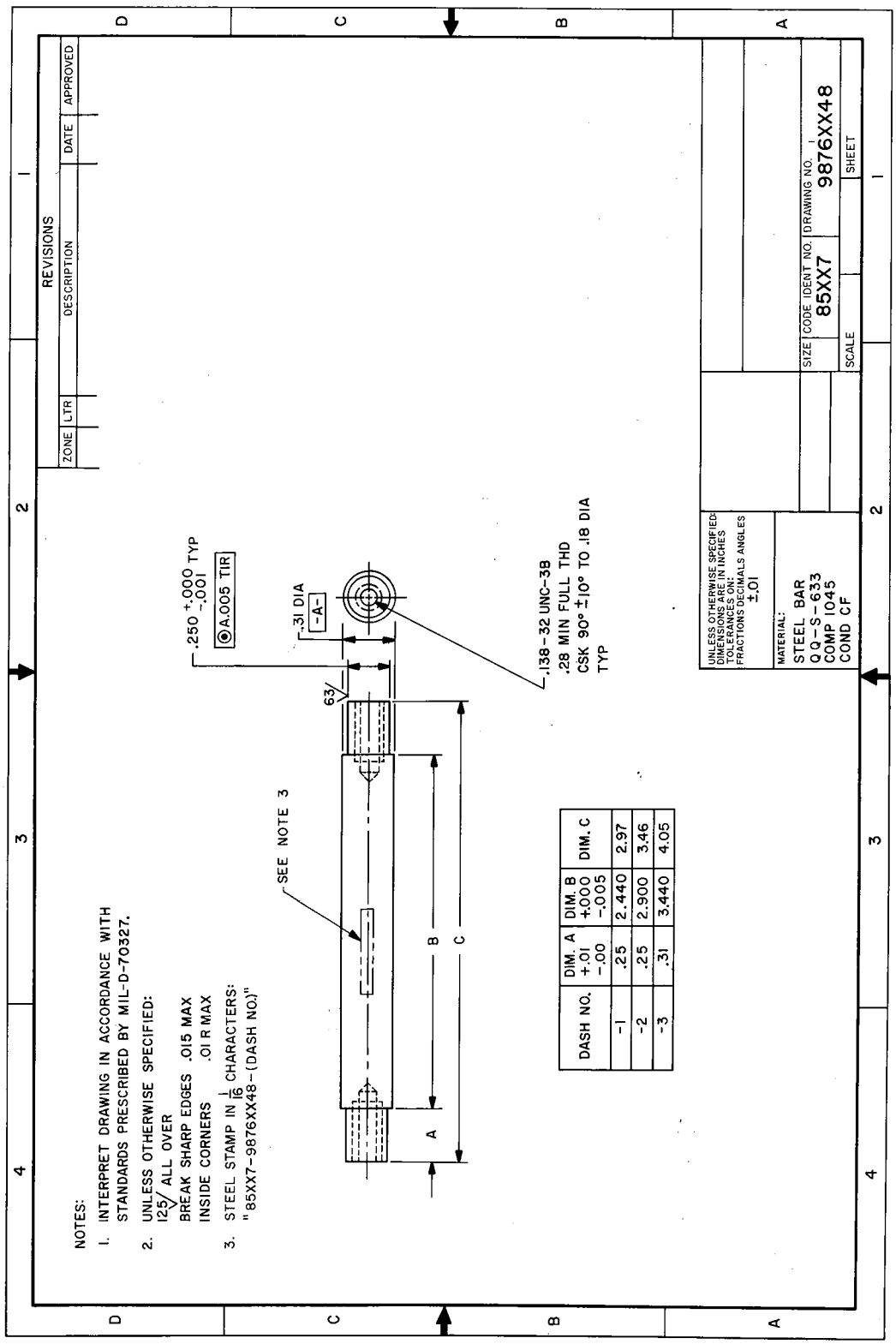


FIGURE 25. Sample Tabulated Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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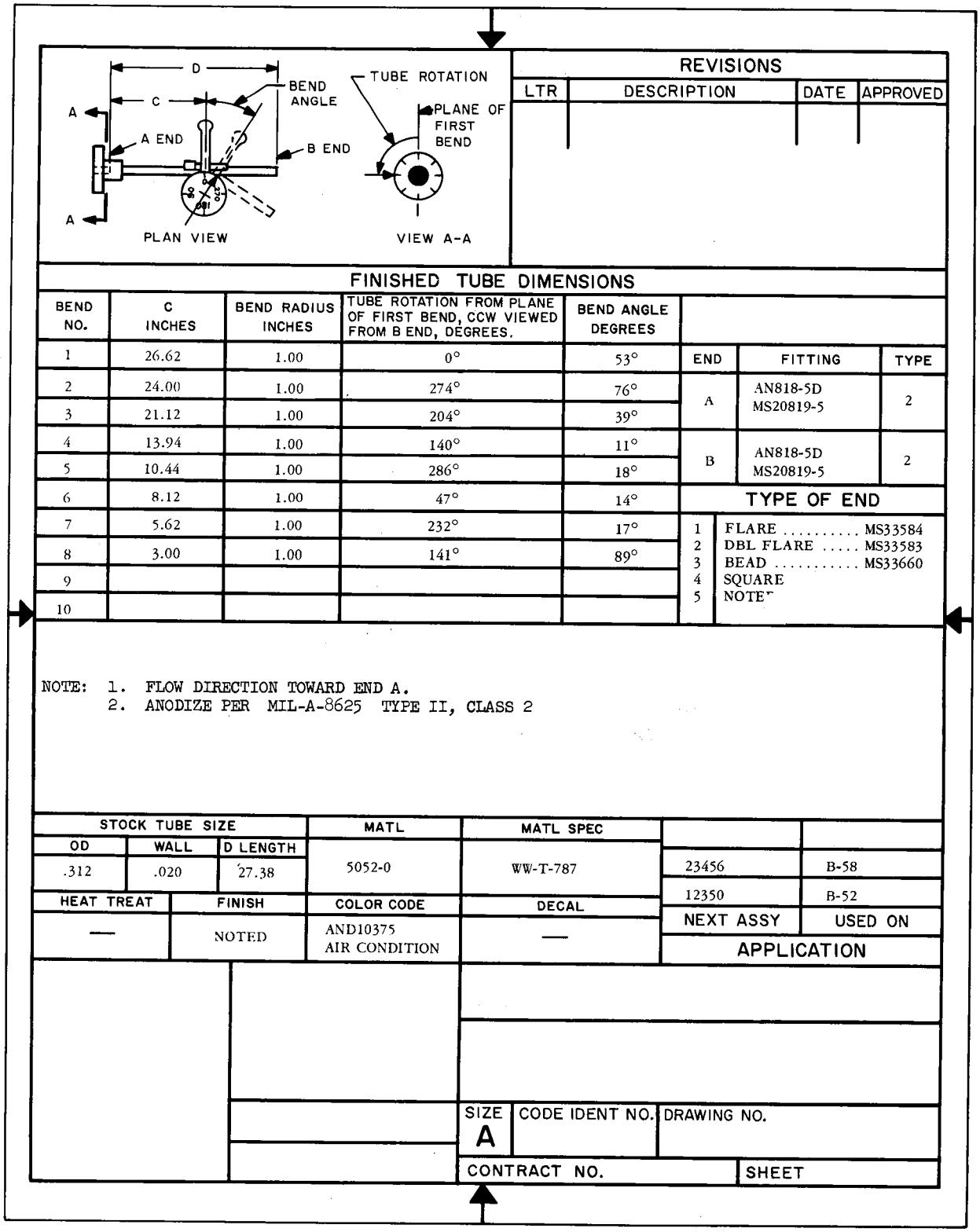


FIGURE 26. Sample Tube Bend Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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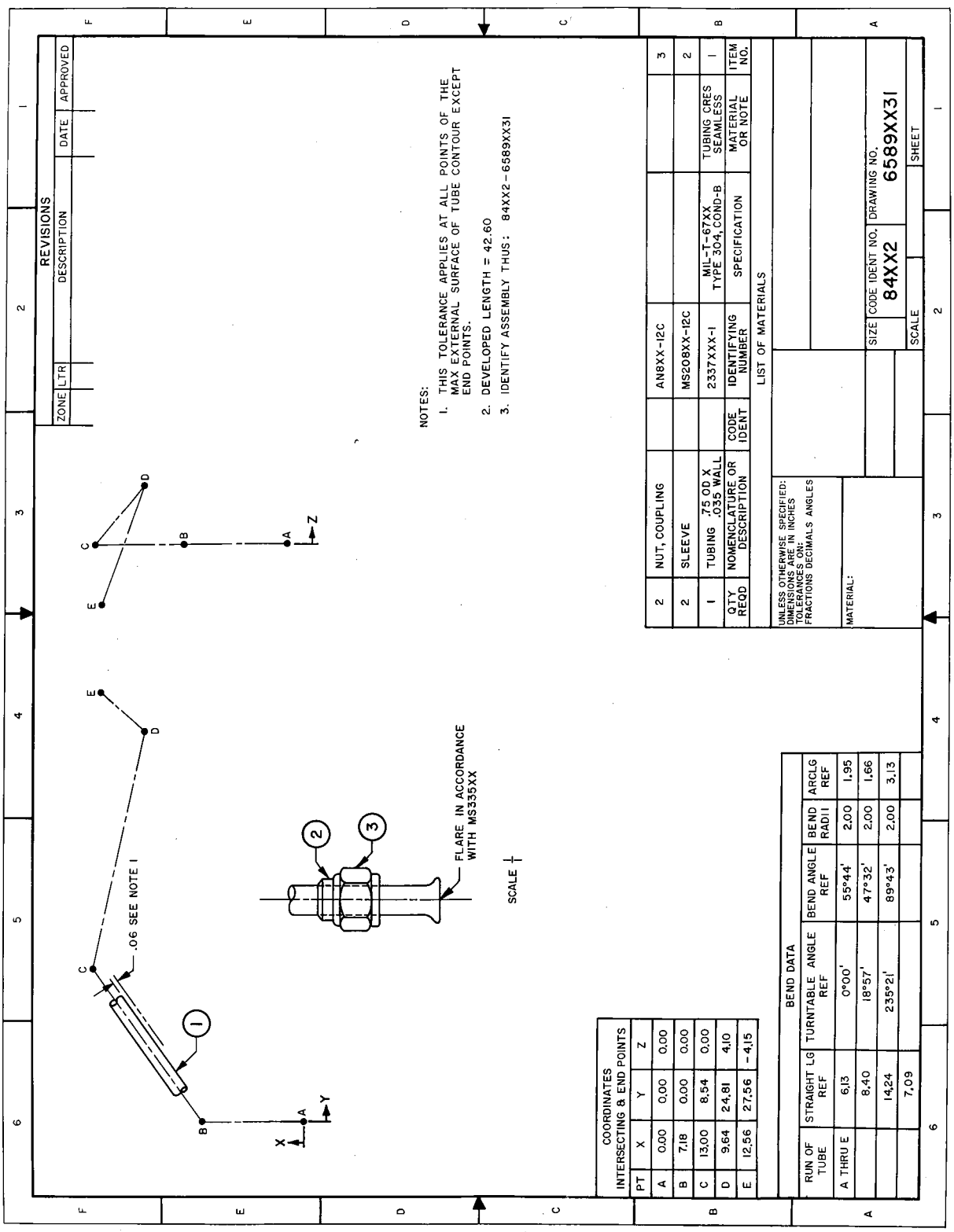


FIGURE 29. Sample Tube Bend Drawing

This sample drawing shall be construed as informational only. It is complete only to the degree necessary to illustrate a particular drawing practice. Actual drawings shall conform to the textual requirements set forth in this standard.

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